Statistics/Data analysis

```
      name:  <unnamed>
       log:  E:\Documents\STATA\cardiovascular.smcl
  log type:  smcl
 opened on:  16 May 2026, 21:05:54

1 .
2 . *import data
3 . import delimited "E:\Documents\STATA\assignment 2_Number 2_2025_2026.csv", varnames(
  > 1)
  (encoding automatically selected: ISO-8859-2)
  (8 vars, 4,699 obs)

4 .
5 . *rename columns
6 .
7 . rename column1 sex

8 . rename column2 sbp

9 . rename column3 dbp

10. rename column4 cholesterol

11. rename column5 age

12. rename column6 persontime

13. rename column7 cv_events

14. rename column8 bmi

15.
16. *Label columns
17.
18. label variable sex "Sex of participant"

19. label variable sbp "Systolic blood pressure"

20. label variable dbp "Diastolic blood pressure"

21. label variable cholesterol "Serum cholesterol"

22. label variable age "Age of participant"

23. label variable persontime "Person time of follow up (days)"

24. label variable cv_events "Number of cardiovascular events"

25. label variable bmi "Body Mass Index"

26.
27.
28. *label values for sex;
29.
30. label define sex_label 1 "Male" 2 "Female"

31. label values sex sex_label

32. tab sex
```

Sex of participant	Freq.	Percent	Cum.
Male	**2,049**	**43.61**	**43.61**
Female	**2,650**	**56.39**	**100.00**
Total	**4,699**	**100.00**	

```
33.
34.
35. *DATA INSPECTION
36.
37. bro

38.
39.
40. *inspection of numeric variable
41. sum sbp dbp cholesterol age persontime cv_events
```

Variable	Obs	Mean	Std. dev.	Min	Max
sbp	**4,699**	**132.7665**	**22.8011**	**80**	**270**
dbp	**4,699**	**82.5416**	**12.73732**	**40**	**148**
cholesterol	**4,666**	**228.2925**	**44.54262**	**115**	**568**
age	**4,699**	**46.04107**	**8.504363**	**30**	**68**
persontime	**4,699**	**8061.313**	**3595.311**	**18**	**11688**
cv_events	**4,699**	**.678655**	**1.293529**	**0**	**11**

```
42.
43. *check extreme outliers in sbp/dbp
44. scatter sbp dbp

45.
46. *inspection of categorical variables
47. tab sex
```

Sex of participant	Freq.	Percent	Cum.
Male	**2,049**	**43.61**	**43.61**
Female	**2,650**	**56.39**	**100.00**
Total	**4,699**	**100.00**	

```
48.
49. *assess missingness
50. misstable summarize
```

				Obs<.		
Variable	Obs=.	Obs>.	Obs<.	Unique values	Min	Max
cholesterol	**33**		**4,666**	248	115	568
bmi	**9**		**4,690**	248	16.2	57.6

```
51. mdesc
```

Variable	Missing	Total	Percent Missing
sex	**0**	**4,699**	**0.00**
sbp	**0**	**4,699**	**0.00**
dbp	**0**	**4,699**	**0.00**
cholesterol	**33**	**4,699**	**0.70**
age	**0**	**4,699**	**0.00**
persontime	**0**	**4,699**	**0.00**
cv_events	**0**	**4,699**	**0.00**
bmi	**9**	**4,699**	**0.19**

```
52.
53. *only cholesterol and bmi have missing data with 0.7% and 0.19%
54.
55.
56.
57. *Explore outcome variables and check for equidispersion
58. tab cv_events
```

Number of cardiovascular events	Freq.	Percent	Cum.
0	**3,070**	**65.33**	**65.33**
1	**933**	**19.86**	**85.19**
2	**296**	**6.30**	**91.49**
3	**192**	**4.09**	**95.57**
4	**88**	**1.87**	**97.45**
5	**58**	**1.23**	**98.68**
6	**24**	**0.51**	**99.19**
7	**18**	**0.38**	**99.57**
8	**12**	**0.26**	**99.83**
9	**4**	**0.09**	**99.91**
11	**4**	**0.09**	**100.00**
Total	**4,699**	**100.00**	

```
59. summarize cv_events, detail
```

Number of cardiovascular events

Percentiles		Smallest		
1%	**0**	**0**		
5%	**0**	**0**		
10%	**0**	**0**	Obs	**4,699**
25%	**0**	**0**	Sum of wgt.	**4,699**
50%	**0**		Mean	**.678655**
		Largest	Std. dev.	**1.293529**
75%	**1**	**11**		
90%	**2**	**11**	Variance	**1.673216**
95%	**3**	**11**	Skewness	**2.914906**
99%	**6**	**11**	Kurtosis	**14.07692**

```
60.
61. *variance> mean
62. *need to check likelihood ratio test for alpha is significant TO DECIDE ON MODEL
63.
64. summarize persontime, detail
```

Person time of follow up (days)

Percentiles		Smallest		
1%	**418**	**18**		
5%	**1449**	**35**		
10%	**2436**	**109**	Obs	**4,699**
25%	**5135**	**147**	Sum of wgt.	**4,699**
50%	**8908**		Mean	**8061.313**
		Largest	Std. dev.	**3595.311**
75%	**11648**	**11688**		
90%	**11688**	**11688**	Variance	**1.29e+07**
95%	**11688**	**11688**	Skewness	**-.5683951**
99%	**11688**	**11688**	Kurtosis	**1.967679**

```
65.
66.
67.
68. *RUN NEGATIVE BINOMIAL MODEL ; to assess likelihood ratio test for alpha
69. nbreg cv_events i.sex sbp dbp cholesterol age bmi, exposure(persontime) irr

  Fitting Poisson model:

  Iteration 0:   log likelihood = -7240.6983
  Iteration 1:   log likelihood = -7240.6948
  Iteration 2:   log likelihood = -7240.6948

  Fitting constant-only model:

  Iteration 0:   log likelihood = -6474.7549
  Iteration 1:   log likelihood = -5956.5974
  Iteration 2:   log likelihood = -5831.8741
  Iteration 3:   log likelihood =  -5821.787
  Iteration 4:   log likelihood = -5821.7579
  Iteration 5:   log likelihood = -5821.7579

  Fitting full model:

  Iteration 0:   log likelihood = -5685.5315
  Iteration 1:   log likelihood = -5667.0254
  Iteration 2:   log likelihood = -5666.6743
  Iteration 3:   log likelihood = -5666.6743

  Negative binomial regression                          Number of obs =  4,658
                                                        LR chi2(6)    =  310.17
  Dispersion: mean                                      Prob > chi2   =  0.0000
  Log likelihood = -5666.6743                           Pseudo R2     =  0.0266

------------------------------------------------------------------------------
   cv_events |        IRR   Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         sex |
      Female |   .6746248   .0490888    -5.41   0.000     .5849584    .7780359
         sbp |   1.004414   .0026356     1.68   0.093     .9992616    1.009593
         dbp |   1.009343   .0046325     2.03   0.043     1.000304    1.018464
 cholesterol |   1.003295   .0008326     3.96   0.000     1.001665    1.004929
         age |   1.055253   .0049112    11.56   0.000     1.045671    1.064922
         bmi |   1.009588   .0091779     1.05   0.294     .9917591    1.027738
       _cons |   1.25e-06   4.14e-07   -40.93   0.000     6.50e-07    2.39e-06
ln(person~e) |          1  (exposure)
-------------+----------------------------------------------------------------
    /lnalpha |   1.321736   .0416339                      1.240135    1.403337
-------------+----------------------------------------------------------------
       alpha |   3.749925   .1561241                      3.456079    4.068754
------------------------------------------------------------------------------
  Note: Estimates are transformed only in the first equation to incidence-rate ratios.
  Note: _cons estimates baseline incidence rate.
  LR test of alpha=0: chibar2(01) = 3148.04              Prob >= chibar2 = 0.000

70. *likelihood ratio test for is significant suggesting that negative binomial is bette
  > r;
71. *however we still need to compare it with other models
72.
73. *CHOOSING MODEL
74. *negative binomial
```

```
75. nbreg cv_events i.sex sbp dbp cholesterol age bmi, exposure(persontime) irr

  Fitting Poisson model:

  Iteration 0:   log likelihood = -7240.6983
  Iteration 1:   log likelihood = -7240.6948
  Iteration 2:   log likelihood = -7240.6948

  Fitting constant-only model:

  Iteration 0:   log likelihood = -6474.7549
  Iteration 1:   log likelihood = -5956.5974
  Iteration 2:   log likelihood = -5831.8741
  Iteration 3:   log likelihood =  -5821.787
  Iteration 4:   log likelihood = -5821.7579
  Iteration 5:   log likelihood = -5821.7579

  Fitting full model:

  Iteration 0:   log likelihood = -5685.5315
  Iteration 1:   log likelihood = -5667.0254
  Iteration 2:   log likelihood = -5666.6743
  Iteration 3:   log likelihood = -5666.6743

  Negative binomial regression                          Number of obs =  4,658
                                                        LR chi2(6)    =  310.17
  Dispersion: mean                                      Prob > chi2   =  0.0000
  Log likelihood = -5666.6743                           Pseudo R2     =  0.0266

------------------------------------------------------------------------------
   cv_events |        IRR   Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
         sex |
      Female |   .6746248   .0490888    -5.41   0.000     .5849584    .7780359
         sbp |   1.004414   .0026356     1.68   0.093     .9992616    1.009593
         dbp |   1.009343   .0046325     2.03   0.043     1.000304    1.018464
 cholesterol |   1.003295   .0008326     3.96   0.000     1.001665    1.004929
         age |   1.055253   .0049112    11.56   0.000     1.045671    1.064922
         bmi |   1.009588   .0091779     1.05   0.294     .9917591    1.027738
       _cons |   1.25e-06   4.14e-07   -40.93   0.000     6.50e-07    2.39e-06
ln(person~e) |          1  (exposure)
-------------+----------------------------------------------------------------
    /lnalpha |   1.321736   .0416339                      1.240135    1.403337
-------------+----------------------------------------------------------------
       alpha |   3.749925   .1561241                      3.456079    4.068754
------------------------------------------------------------------------------
  Note: Estimates are transformed only in the first equation to incidence-rate ratios.
  Note: _cons estimates baseline incidence rate.
  LR test of alpha=0: chibar2(01) = 3148.04              Prob >= chibar2 = 0.000

76. estimates store nbreg

77.
78. *poisson model
79. poisson cv_events i.sex  sbp dbp cholesterol age bmi, exposure(persontime) irr

  Iteration 0:   log likelihood = -7240.6983
  Iteration 1:   log likelihood = -7240.6948
  Iteration 2:   log likelihood = -7240.6948

  Poisson regression                                    Number of obs =   4,658
                                                        LR chi2(6)    = 1075.56
                                                        Prob > chi2   =  0.0000
  Log likelihood = -7240.6948                           Pseudo R2     =  0.0691
```

cv_events	IRR	Std. err.	z	P>|z|	[95% conf.	interval]
sex						
Female	.6675981	.0243717	-11.07	0.000	.6214993	.7171161
sbp	1.006415	.0012629	5.10	0.000	1.003943	1.008894
dbp	1.004989	.0022794	2.19	0.028	1.000532	1.009467
cholesterol	1.002326	.0004138	5.63	0.000	1.001515	1.003137
age	1.049976	.0024626	20.79	0.000	1.045161	1.054814
bmi	1.011269	.004663	2.43	0.015	1.002171	1.02045
_cons	1.33e-06	2.28e-07	-79.17	0.000	9.54e-07	1.86e-06
ln(person~e)	1	(exposure)				

Note: **_cons** estimates baseline incidence rate.

```
80. estimates store poisson

81.
82. *Zero-Inflated Poisson Model
83. zip cv_events i.sex sbp dbp cholesterol age bmi, inflate(i.sex sbp dbp cholesterol a
  > ge bmi) exposure(persontime)
  (41 missing values generated)

  Fitting constant-only model:

  Iteration 0:   log likelihood = -6501391.4
  Iteration 1:   log likelihood = -12306.751
  Iteration 2:   log likelihood =  -11123.77  (not concave)
  Iteration 3:   log likelihood = -7923.1916
  Iteration 4:   log likelihood = -7842.1033  (not concave)
  Iteration 5:   log likelihood = -7780.6996
  Iteration 6:   log likelihood =  -7513.326  (not concave)
  Iteration 7:   log likelihood = -7309.3732  (not concave)
  Iteration 8:   log likelihood = -6821.1024
  Iteration 9:   log likelihood = -6148.5663
  Iteration 10:  log likelihood = -6075.8075
  Iteration 11:  log likelihood = -6075.5591
  Iteration 12:  log likelihood = -6075.5591

  Fitting full model:

  Iteration 0:   log likelihood = -6075.5591
  Iteration 1:   log likelihood = -6035.2714
  Iteration 2:   log likelihood = -6034.6269
  Iteration 3:   log likelihood = -6034.6266

  Zero-inflated Poisson regression                        Number of obs =  4,658
  Inflation model: logit                                  Nonzero obs   =  1,614
                                                          Zero obs      =  3,044
                                                          LR chi2(6)    =  81.86
  Log likelihood = -6034.627                              Prob > chi2   = 0.0000
```

cv_events	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
cv_events						
sex						
Female	.0318538	.0428507	0.74	0.457	-.0521319	.1158396
sbp	-.0009767	.0014662	-0.67	0.505	-.0038505	.001897
dbp	.0052493	.0026482	1.98	0.047	.000059	.0104395
cholesterol	-.0003703	.0005007	-0.74	0.460	-.0013516	.000611
age	.0212858	.0027014	7.88	0.000	.0159912	.0265803
bmi	-.0026702	.0052972	-0.50	0.614	-.0130525	.007712
_cons	-9.338053	.2097091	-44.53	0.000	-9.749076	-8.927031
ln(person~e)	1	(exposure)				
inflate						
sex						
Female	.626338	.0737294	8.50	0.000	.4818311	.770845
sbp	-.0120699	.0028002	-4.31	0.000	-.0175581	-.0065817
dbp	-.0011777	.004784	-0.25	0.806	-.0105541	.0081987
cholesterol	-.0039543	.0008651	-4.57	0.000	-.0056498	-.0022587

age	-.0415112	.0047377	-8.76	0.000	-.0507968	-.0322256
bmi	-.0187869	.0096397	-1.95	0.051	-.0376804	.0001066
_cons	5.090136	.350521	14.52	0.000	4.403127	5.777144

```
84.
85. **remove non significant predictors in inflated
86. *drop dpb
87. zip cv_events i.sex sbp dbp cholesterol age bmi, inflate(i.sex sbp cholesterol age b
  > mi) exposure(persontime)
  (41 missing values generated)

  Fitting constant-only model:

  Iteration 0:   log likelihood = -6501391.4
  Iteration 1:   log likelihood = -12317.345
  Iteration 2:   log likelihood = -11137.976  (not concave)
  Iteration 3:   log likelihood = -7930.0467
  Iteration 4:   log likelihood = -7370.1306  (not concave)
  Iteration 5:   log likelihood = -7015.5534
  Iteration 6:   log likelihood = -6218.1623
  Iteration 7:   log likelihood = -6076.6561
  Iteration 8:   log likelihood = -6075.7399
  Iteration 9:   log likelihood = -6075.7397

  Fitting full model:

  Iteration 0:   log likelihood = -6075.7397
  Iteration 1:   log likelihood = -6035.2999
  Iteration 2:   log likelihood = -6034.6572
  Iteration 3:   log likelihood = -6034.6569

  Zero-inflated Poisson regression                        Number of obs =  4,658
  Inflation model: logit                                  Nonzero obs   =  1,614
                                                          Zero obs      =  3,044
                                                          LR chi2(6)    =  82.17
  Log likelihood = -6034.657                              Prob > chi2   = 0.0000
```

cv_events	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
cv_events						
sex						
Female	.0322131	.042828	0.75	0.452	-.0517281	.1161544
sbp	-.0010464	.0014395	-0.73	0.467	-.0038677	.001775
dbp	.0054082	.0025689	2.11	0.035	.0003733	.0104432
cholesterol	-.0003736	.0005006	-0.75	0.455	-.0013547	.0006075
age	.0213112	.0027001	7.89	0.000	.016019	.0266033
bmi	-.0027261	.0052928	-0.52	0.607	-.0130999	.0076476
_cons	-9.341292	.2093958	-44.61	0.000	-9.7517	-8.930884
ln(person~e)	1	(exposure)				
inflate						
sex						
Female	.628726	.0731044	8.60	0.000	.4854439	.772008
sbp	-.0125861	.00186	-6.77	0.000	-.0162316	-.0089406
cholesterol	-.0039668	.0008638	-4.59	0.000	-.0056597	-.0022738
age	-.0413628	.0046997	-8.80	0.000	-.0505741	-.0321515
bmi	-.019257	.0094544	-2.04	0.042	-.0377873	-.0007268
_cons	5.067778	.3386131	14.97	0.000	4.404109	5.731448

```
88.  estimates store zpoisson

89.
90.
91. *Zero-Inflated Negative Binomial model
92. zinb cv_events i.sex sbp dbp cholesterol age bmi, inflate(i.sex sbp dbp cholesterol
  > age bmi) exposure(persontime)

  Fitting constant-only model:

  Iteration 0:   log likelihood =  -6408.826  (not concave)
  Iteration 1:   log likelihood = -5829.1535  (not concave)
  Iteration 2:   log likelihood = -5765.2956  (not concave)
  Iteration 3:   log likelihood = -5750.8509  (not concave)
  Iteration 4:   log likelihood = -5743.6447  (not concave)
  Iteration 5:   log likelihood = -5723.2756  (not concave)
  Iteration 6:   log likelihood =  -5709.385  (not concave)
  Iteration 7:   log likelihood = -5705.2117  (not concave)
  Iteration 8:   log likelihood = -5699.4836  (not concave)
  Iteration 9:   log likelihood = -5695.6879
  Iteration 10:  log likelihood = -5684.2184
  Iteration 11:  log likelihood = -5637.7295
  Iteration 12:  log likelihood = -5633.0091
  Iteration 13:  log likelihood = -5632.6952
  Iteration 14:  log likelihood = -5632.6951

  Fitting full model:

  Iteration 0:   log likelihood = -5632.6951
  Iteration 1:   log likelihood = -5606.4369
  Iteration 2:   log likelihood = -5605.7223
  Iteration 3:   log likelihood =  -5605.715
  Iteration 4:   log likelihood =  -5605.715

  Zero-inflated negative binomial regression              Number of obs =  4,658
  Inflation model: logit                                  Nonzero obs   =  1,614
                                                          Zero obs      =  3,044
                                                          LR chi2(6)    =  53.96
  Log likelihood = -5605.715                              Prob > chi2   = 0.0000

------------------------------------------------------------------------------
   cv_events | Coefficient  Std. err.      z    P>|z|     [95% conf. interval]
-------------+----------------------------------------------------------------
cv_events    |
         sex |
      Female |  -.0651561    .086659    -0.75   0.452    -.2350047    .1046925
         sbp |   .0001879   .0027637     0.07   0.946     -.005229    .0056047
         dbp |   .0083953   .0049917     1.68   0.093    -.0013882    .0181788
 cholesterol |   .0003678   .0009136     0.40   0.687    -.0014229    .0021584
         age |   .0303428    .005374     5.65   0.000       .01981    .0408757
         bmi |  -.0007545   .0097566    -0.08   0.938    -.0198771    .0183682
       _cons |  -10.69444   .4164003   -25.68   0.000    -11.51057   -9.878312
ln(person~e) |          1  (exposure)
-------------+----------------------------------------------------------------
inflate      |
         sex |
      Female |   .9405915   .1809496     5.20   0.000     .5859368    1.295246
         sbp |  -.0227536    .007094    -3.21   0.001    -.0366576   -.0088497
         dbp |   .0041214   .0112311     0.37   0.714    -.0178912    .0261339
 cholesterol |  -.0075431    .002159    -3.49   0.000    -.0117747   -.0033115
         age |  -.0633564   .0119218    -5.31   0.000    -.0867226   -.0399902
         bmi |  -.0271251    .020779    -1.31   0.192    -.0678511    .0136009
       _cons |   6.520331   .8581005     7.60   0.000     4.838485    8.202177
-------------+----------------------------------------------------------------
    /lnalpha |   .8035613   .0854523     9.40   0.000     .6360778    .9710448
-------------+----------------------------------------------------------------
       alpha |   2.233481   .1908561                      1.889057    2.640702
------------------------------------------------------------------------------
```

```
93.
94. **drop dbp in inflated
95. zinb cv_events i.sex sbp dbp cholesterol age bmi, inflate(i.sex sbp cholesterol age
  > bmi) exposure(persontime)

  Fitting constant-only model:

  Iteration 0:   log likelihood =  -6408.826  (not concave)
  Iteration 1:   log likelihood = -5829.3975  (not concave)
  Iteration 2:   log likelihood = -5765.7949  (not concave)
  Iteration 3:   log likelihood = -5751.1908  (not concave)
  Iteration 4:   log likelihood = -5743.8927  (not concave)
  Iteration 5:   log likelihood = -5728.0639  (not concave)
  Iteration 6:   log likelihood = -5722.2597  (not concave)
  Iteration 7:   log likelihood =  -5717.043  (not concave)
  Iteration 8:   log likelihood =  -5712.773  (not concave)
  Iteration 9:   log likelihood = -5708.4112  (not concave)
  Iteration 10:  log likelihood = -5704.4902  (not concave)
  Iteration 11:  log likelihood =  -5700.663
  Iteration 12:  log likelihood = -5675.1838
  Iteration 13:  log likelihood = -5636.8399
  Iteration 14:  log likelihood = -5632.9987
  Iteration 15:  log likelihood = -5632.7884
  Iteration 16:  log likelihood = -5632.7878
  Iteration 17:  log likelihood = -5632.7878

  Fitting full model:

  Iteration 0:   log likelihood = -5632.7878
  Iteration 1:   log likelihood = -5606.4851
  Iteration 2:   log likelihood = -5605.7892
  Iteration 3:   log likelihood = -5605.7824
  Iteration 4:   log likelihood = -5605.7824

  Zero-inflated negative binomial regression              Number of obs =  4,658
  Inflation model: logit                                  Nonzero obs   =  1,614
                                                          Zero obs      =  3,044
                                                          LR chi2(6)    =  54.01
  Log likelihood = -5605.782                              Prob > chi2   = 0.0000
```

cv_events	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
cv_events						
sex						
Female	-.0671452	.0864942	-0.78	0.438	-.2366708	.1023804
sbp	.0005694	.0025655	0.22	0.824	-.0044589	.0055977
dbp	.0075347	.0044086	1.71	0.087	-.001106	.0161754
cholesterol	.0003974	.0009109	0.44	0.663	-.0013879	.0021828
age	.0300916	.0053269	5.65	0.000	.019651	.0405322
bmi	-.0005089	.0097402	-0.05	0.958	-.0195992	.0185815
_cons	-10.67464	.4127807	-25.86	0.000	-11.48367	-9.865601
ln(person~e)	1	(exposure)				
inflate						
sex						
Female	.9358061	.1807326	5.18	0.000	.5815768	1.290035
sbp	-.0208663	.004814	-4.33	0.000	-.0303017	-.011431
cholesterol	-.0074216	.0021276	-3.49	0.000	-.0115916	-.0032516
age	-.0641874	.0117412	-5.47	0.000	-.0871998	-.041175
bmi	-.0257681	.0204868	-1.26	0.208	-.0659216	.0143854
_cons	6.587863	.838865	7.85	0.000	4.943718	8.232009
/lnalpha	.8047577	.0854576	9.42	0.000	.6372639	.9722515
alpha	2.236155	.1910964			1.891299	2.64389

```
96.
97. **drop bmi
98. zinb cv_events i.sex sbp dbp cholesterol age bmi, inflate(i.sex sbp cholesterol age)
  >  exposure(persontime)

  Fitting constant-only model:

  Iteration 0:   log likelihood =  -6408.826  (not concave)
  Iteration 1:   log likelihood = -5828.8948  (not concave)
  Iteration 2:   log likelihood = -5765.1166  (not concave)
  Iteration 3:   log likelihood = -5750.3108  (not concave)
  Iteration 4:   log likelihood = -5742.9957  (not concave)
  Iteration 5:   log likelihood = -5729.1703  (not concave)
  Iteration 6:   log likelihood =  -5722.544  (not concave)
  Iteration 7:   log likelihood = -5648.6204
  Iteration 8:   log likelihood =  -5636.114
  Iteration 9:   log likelihood = -5634.0555
  Iteration 10:  log likelihood = -5634.0476
  Iteration 11:  log likelihood = -5634.0476

  Fitting full model:

  Iteration 0:   log likelihood = -5634.0476
  Iteration 1:   log likelihood = -5607.3201
  Iteration 2:   log likelihood = -5606.6037
  Iteration 3:   log likelihood = -5606.5964
  Iteration 4:   log likelihood = -5606.5964

  Zero-inflated negative binomial regression            Number of obs =  4,658
  Inflation model: logit                                Nonzero obs   =  1,614
                                                        Zero obs      =  3,044
                                                        LR chi2(6)    =  54.90
  Log likelihood = -5606.596                            Prob > chi2   = 0.0000
```

cv_events	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
cv_events						
sex						
Female	-.0675711	.0865404	-0.78	0.435	-.2371871	.1020449
sbp	.0003117	.0025566	0.12	0.903	-.0046991	.0053224
dbp	.0075713	.0044091	1.72	0.086	-.0010704	.0162131
cholesterol	.000344	.0009088	0.38	0.705	-.0014372	.0021251
age	.0301856	.0053324	5.66	0.000	.0197343	.0406369
bmi	.0048426	.0088309	0.55	0.583	-.0124656	.0221509
_cons	-10.77852	.4067777	-26.50	0.000	-11.57579	-9.981249
ln(person~e)	1	(exposure)				
inflate						
sex						
Female	.9603003	.1819917	5.28	0.000	.6036032	1.316997
sbp	-.0227638	.0047376	-4.80	0.000	-.0320493	-.0134783
cholesterol	-.0077602	.0021459	-3.62	0.000	-.0119662	-.0035543
age	-.0650355	.0118164	-5.50	0.000	-.0881952	-.0418757
_cons	6.257369	.7990061	7.83	0.000	4.691346	7.823392
/lnalpha	.8132231	.0849657	9.57	0.000	.6466933	.9797529
alpha	2.255165	.1916118			1.909217	2.663798

```
99.
100 estimates store znegbin

101
102
103
104
105 *wont generate zero truncated models as the cardiovascular events have numerous zero
  > es
106
107 *now we compare the  models
108 lrtest ( poisson nbreg) ( zpoisson znegbin), stats force

  Likelihood-ratio test
  Assumption: (poisson, nbreg) nested within (zpoisson, znegbin)

  LR chi2(11) = 2532.23
  Prob > chi2 =  0.0000

  Akaike's information criterion and Bayesian information criterion

┌──────────────────────────────────────────────────────────────────────────────┐
│       Model |          N   ll(null)  ll(model)      df        AIC        BIC │
├─────────────┼────────────────────────────────────────────────────────────────┤
│     poisson |      4,658  -7778.475  -7240.695       7   14495.39   14540.51 │
│       nbreg |      4,658  -5821.758  -5666.674       8   11349.35   11400.92 │
│    zpoisson |      4,658   -6075.74  -6034.657      13   12095.31   12179.12 │
│     znegbin |      4,658  -5634.048  -5606.596      13   11239.19      11323 │
└──────────────────────────────────────────────────────────────────────────────┘
  Note: BIC uses N = number of observations. See [R] BIC note.

109 *AIC/BIC for zero inflated  negative binomial are smallest ; hence zero inflated neg
  > ative binomial model is best for this analysis;
110
111 *BEFORE VARIABLE SELECTION WE ASSESS IF BP IS KEPT AS CONTINOUS OR CATEGORICAL AND T
  > HEN BMI
112
113 *BP
114 *BLOOD PRESSURE
115 *1.sbp as is
116 *2.dbp as is
117 *3.binary sbp (high sbp; 140mg and above); yes(1)/no(0)
118 *4.binary dbp (high dpb; 100mh and above); yes(1)/no(0)
119 *5.hypertension(systolic 140mmHg and above OR diastolic 100mmHg and above); 1(yes) a
  > nd 0(no)
120
121 *3 binary sbp
122 gen high_sbp = 0

123 replace high_sbp = 1 if sbp >= 140
  (1,470 real changes made)

124 replace high_sbp = . if sbp == .
  (0 real changes made)

125 label variable high_sbp "High Systolic BP 140+ (1=Yes, 0=No)"

126 label define yesnolabel 0"No" 1"Yes"

127 label values high_sbp yesnolabel
```

```
128
129 *4. binary dbp
130 gen high_dbp = 0

131 replace high_dbp =1 if dbp >= 100
  (476 real changes made)

132 replace high_dbp = . if dbp == .
  (0 real changes made)

133 label variable high_dbp "High Diastolic BP 100+ (1=Yes, 0=No)"

134 label values high_dbp yesnolabel

135
136 *5 hypertension
137 gen hypertension = 0

138 replace hypertension =1 if sbp >= 140 | dbp >= 100
  (1,492 real changes made)

139 replace hypertension = . if sbp ==. & dbp == .
  (0 real changes made)

140 label variable hypertension "Hypertension"

141 label values hypertension yesnolabel

142
143 *check
144 tab high_sbp
```

High Systolic BP 140+ (1=Yes, 0=No)	Freq.	Percent	Cum.
No	**3,229**	**68.72**	**68.72**
Yes	**1,470**	**31.28**	**100.00**
Total	**4,699**	**100.00**	

```
145 tab high_dbp
```

High Diastolic BP 100+ (1=Yes, 0=No)	Freq.	Percent	Cum.
No	**4,223**	**89.87**	**89.87**
Yes	**476**	**10.13**	**100.00**
Total	**4,699**	**100.00**	

```
146 tab hypertension
```

Hypertension	Freq.	Percent	Cum.
No	**3,207**	**68.25**	**68.25**
Yes	**1,492**	**31.75**	**100.00**
Total	**4,699**	**100.00**	

```
147
148 *BMI
149 gen obese_ow = 0

150 replace obese_ow =1 if bmi >=25
  (2,476 real changes made)

151 replace obese_ow = . if bmi==.
  (9 real changes made, 9 to missing)

152 label variable obese_ow "Overweight or Obese"

153 label values obese_ow yesnolabel

154
155 tab obese_ow
```

Overweight or Obese	Freq.	Percent	Cum.
No	**2,223**	**47.40**	**47.40**
Yes	**2,467**	**52.60**	**100.00**
Total	**4,690**	**100.00**	

```
156
157
158 *NOW WE CREATE DIFFERENT MODELS WITH THE DIFFERENT COMBINATIONS OF BP
159 *The models;
160 *1. BP AS IS
161 zinb cv_events i.sex sbp dbp cholesterol age bmi, inflate(i.sex sbp cholesterol age)
  >  exposure(persontime)

  Fitting constant-only model:

  Iteration 0:   log likelihood =  -6408.826  (not concave)
  Iteration 1:   log likelihood = -5828.8948  (not concave)
  Iteration 2:   log likelihood = -5765.1166  (not concave)
  Iteration 3:   log likelihood = -5750.3108  (not concave)
  Iteration 4:   log likelihood = -5742.9957  (not concave)
  Iteration 5:   log likelihood = -5729.1703  (not concave)
  Iteration 6:   log likelihood =  -5722.544  (not concave)
  Iteration 7:   log likelihood = -5648.6204
  Iteration 8:   log likelihood =  -5636.114
  Iteration 9:   log likelihood = -5634.0555
  Iteration 10:  log likelihood = -5634.0476
  Iteration 11:  log likelihood = -5634.0476

  Fitting full model:

  Iteration 0:   log likelihood = -5634.0476
  Iteration 1:   log likelihood = -5607.3201
  Iteration 2:   log likelihood = -5606.6037
  Iteration 3:   log likelihood = -5606.5964
  Iteration 4:   log likelihood = -5606.5964

  Zero-inflated negative binomial regression            Number of obs =  4,658
  Inflation model: logit                                Nonzero obs   =  1,614
                                                        Zero obs      =  3,044
                                                        LR chi2(6)    =  54.90
  Log likelihood = -5606.596                            Prob > chi2   = 0.0000
```

cv_events	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
cv_events						
sex						
Female	-.0675711	.0865404	-0.78	0.435	-.2371871	.1020449
sbp	.0003117	.0025566	0.12	0.903	-.0046991	.0053224
dbp	.0075713	.0044091	1.72	0.086	-.0010704	.0162131
cholesterol	.000344	.0009088	0.38	0.705	-.0014372	.0021251
age	.0301856	.0053324	5.66	0.000	.0197343	.0406369
bmi	.0048426	.0088309	0.55	0.583	-.0124656	.0221509
_cons	-10.77852	.4067777	-26.50	0.000	-11.57579	-9.981249
ln(person~e)	1	(exposure)				
inflate						
sex						
Female	.9603003	.1819917	5.28	0.000	.6036032	1.316997
sbp	-.0227638	.0047376	-4.80	0.000	-.0320493	-.0134783
cholesterol	-.0077602	.0021459	-3.62	0.000	-.0119662	-.0035543
age	-.0650355	.0118164	-5.50	0.000	-.0881952	-.0418757
_cons	6.257369	.7990061	7.83	0.000	4.691346	7.823392
/lnalpha	.8132231	.0849657	9.57	0.000	.6466933	.9797529
alpha	2.255165	.1916118			1.909217	2.663798

```
162 estimates store bp_as_is

163
164 *2. spb binary categorical and dbp continous
165
166 zinb cv_events i.sex i.high_sbp dbp cholesterol age bmi, inflate(i.sex i.high_sbp ch
  > olesterol age) exposure(persontime)

  Fitting constant-only model:

  Iteration 0:   log likelihood =  -6408.826  (not concave)
  Iteration 1:   log likelihood = -5809.3088  (not concave)
  Iteration 2:   log likelihood = -5714.9908  (not concave)
  Iteration 3:   log likelihood =  -5703.216
  Iteration 4:   log likelihood =  -5693.667
  Iteration 5:   log likelihood = -5646.1272
  Iteration 6:   log likelihood = -5640.4959
  Iteration 7:   log likelihood = -5639.9508
  Iteration 8:   log likelihood =  -5639.946
  Iteration 9:   log likelihood =  -5639.946

  Fitting full model:

  Iteration 0:   log likelihood =  -5639.946
  Iteration 1:   log likelihood = -5609.8644
  Iteration 2:   log likelihood = -5609.0184
  Iteration 3:   log likelihood = -5609.0074
  Iteration 4:   log likelihood = -5609.0074

  Zero-inflated negative binomial regression              Number of obs =  4,658
  Inflation model: logit                                  Nonzero obs   =  1,614
                                                          Zero obs      =  3,044
                                                          LR chi2(6)    =  61.88
  Log likelihood = -5609.007                              Prob > chi2   = 0.0000
```

cv_events	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
cv_events						
sex						
Female	**-.0528178**	**.085839**	**-0.62**	**0.538**	**-.2210592**	**.1154236**
high_sbp						
Yes	**-.1053046**	**.1092982**	**-0.96**	**0.335**	**-.3195252**	**.108916**
dbp	**.0114065**	**.003484**	**3.27**	**0.001**	**.0045779**	**.0182351**
cholesterol	**.0003594**	**.000913**	**0.39**	**0.694**	**-.0014301**	**.0021489**
age	**.0323713**	**.005299**	**6.11**	**0.000**	**.0219854**	**.0427572**
bmi	**.0051979**	**.0088734**	**0.59**	**0.558**	**-.0121937**	**.0225895**
_cons	**-11.16544**	**.4500848**	**-24.81**	**0.000**	**-12.04759**	**-10.28329**
ln(person~e)	**1**	(exposure)				
inflate						
sex						
Female	**1.065658**	**.1932137**	**5.52**	**0.000**	**.6869661**	**1.44435**
high_sbp						
Yes	**-1.172669**	**.3210174**	**-3.65**	**0.000**	**-1.801852**	**-.5434868**
cholesterol	**-.0085773**	**.0022296**	**-3.85**	**0.000**	**-.0129473**	**-.0042073**
age	**-.0679092**	**.0121317**	**-5.60**	**0.000**	**-.0916869**	**-.0441316**
_cons	**3.743486**	**.6940309**	**5.39**	**0.000**	**2.38321**	**5.103761**
/lnalpha	**.8501101**	**.0822927**	**10.33**	**0.000**	**.6888193**	**1.011401**
alpha	**2.339904**	**.1925571**			**1.991363**	**2.74945**

```
167 estimates store cat_sbp

168
169 *3. dbp binary categorical and spb continous
170 zinb cv_events i.sex sbp i.high_dbp cholesterol age bmi, inflate(i.sex sbp cholester
  > ol age) exposure(persontime)

  Fitting constant-only model:

  Iteration 0:   log likelihood =  -6408.826  (not concave)
  Iteration 1:   log likelihood = -5828.8948  (not concave)
  Iteration 2:   log likelihood = -5765.1166  (not concave)
  Iteration 3:   log likelihood = -5750.3108  (not concave)
  Iteration 4:   log likelihood = -5742.9957  (not concave)
  Iteration 5:   log likelihood = -5729.1703  (not concave)
  Iteration 6:   log likelihood =  -5722.544  (not concave)
  Iteration 7:   log likelihood = -5648.6204
  Iteration 8:   log likelihood =  -5636.114
  Iteration 9:   log likelihood = -5634.0555
  Iteration 10:  log likelihood = -5634.0476
  Iteration 11:  log likelihood = -5634.0476

  Fitting full model:

  Iteration 0:   log likelihood = -5634.0476
  Iteration 1:   log likelihood = -5606.8014
  Iteration 2:   log likelihood = -5606.0658
  Iteration 3:   log likelihood =  -5606.058
  Iteration 4:   log likelihood = -5606.0579

  Zero-inflated negative binomial regression          Number of obs =  4,658
  Inflation model: logit                              Nonzero obs   =  1,614
                                                      Zero obs      =  3,044
                                                      LR chi2(6)    =  55.98
  Log likelihood = -5606.058                          Prob > chi2   = 0.0000
```

cv_events	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
cv_events						
sex						
Female	-.0811715	.0855654	-0.95	0.343	-.2488766	.0865335
sbp	.0012645	.0020715	0.61	0.542	-.0027956	.0053245
high_dbp						
Yes	.2554634	.1280647	1.99	0.046	.0044613	.5064656
cholesterol	.0004767	.0009077	0.53	0.599	-.0013024	.0022558
age	.0296087	.0052668	5.62	0.000	.019286	.0399315
bmi	.0067369	.0086987	0.77	0.439	-.0103122	.0237861
_cons	-10.35211	.4060612	-25.49	0.000	-11.14798	-9.556249
ln(person~e)	1	(exposure)				
inflate						
sex						
Female	.9555056	.1819343	5.25	0.000	.5989209	1.31209
sbp	-.0236899	.0048955	-4.84	0.000	-.033285	-.0140948
cholesterol	-.0077326	.0021511	-3.59	0.000	-.0119486	-.0035166
age	-.0658073	.0119001	-5.53	0.000	-.0891311	-.0424836
_cons	6.395834	.8150403	7.85	0.000	4.798384	7.993283
/lnalpha	.8158388	.0845813	9.65	0.000	.6500625	.9816151
alpha	2.261071	.1912443			1.915661	2.668763

```
171 estimates store cat_dbp

172
173 *4 both sbp and dbp as binary categorical
174 zinb cv_events i.sex i.high_sbp i.high_dbp cholesterol age bmi, inflate(i.sex i.high
  > _sbp cholesterol age) exposure(persontime)

  Fitting constant-only model:

  Iteration 0:   log likelihood =  -6408.826  (not concave)
  Iteration 1:   log likelihood = -5809.3088  (not concave)
  Iteration 2:   log likelihood = -5714.9908  (not concave)
  Iteration 3:   log likelihood =  -5703.216
  Iteration 4:   log likelihood =  -5693.667
  Iteration 5:   log likelihood = -5646.1272
  Iteration 6:   log likelihood = -5640.4959
  Iteration 7:   log likelihood = -5639.9508
  Iteration 8:   log likelihood =  -5639.946
  Iteration 9:   log likelihood =  -5639.946

  Fitting full model:

  Iteration 0:   log likelihood =  -5639.946
  Iteration 1:   log likelihood = -5611.0109
  Iteration 2:   log likelihood = -5610.1985
  Iteration 3:   log likelihood = -5610.1873
  Iteration 4:   log likelihood = -5610.1872

  Zero-inflated negative binomial regression              Number of obs =  4,658
  Inflation model: logit                                  Nonzero obs   =  1,614
                                                          Zero obs      =  3,044
                                                          LR chi2(6)    =  59.52
  Log likelihood = -5610.187                              Prob > chi2   = 0.0000
```

cv_events	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
cv_events						
sex						
Female	-.0726353	.0853619	-0.85	0.395	-.2399415	.0946708
high_sbp						
Yes	-.0198382	.0990114	-0.20	0.841	-.2138969	.1742205
high_dbp						
Yes	.3321674	.1155432	2.87	0.004	.1057068	.5586279
cholesterol	.0005399	.0009123	0.59	0.554	-.0012482	.002328
age	.0316282	.0053085	5.96	0.000	.0212237	.0420326
bmi	.009525	.0086669	1.10	0.272	-.0074618	.0265117
_cons	-10.39025	.3920861	-26.50	0.000	-11.15872	-9.621772
ln(person~e)	1	(exposure)				
inflate						
sex						
Female	1.067122	.1939308	5.50	0.000	.6870249	1.44722
high_sbp						
Yes	-1.163381	.3225388	-3.61	0.000	-1.795545	-.5312161
cholesterol	-.0086205	.0022414	-3.85	0.000	-.0130136	-.0042274
age	-.0698052	.0122834	-5.68	0.000	-.0938802	-.0457301
_cons	3.827673	.7003004	5.47	0.000	2.455109	5.200236
/lnalpha	.852633	.0821686	10.38	0.000	.6915856	1.01368
alpha	2.345815	.1927523			1.996879	2.755725

```
175 estimates store both_cat

176
177 *5 hypertension coded
178 zinb cv_events i.sex i.hypertension cholesterol age bmi, inflate(i.sex i.hypertensio
  > n cholesterol age) exposure(persontime)

  Fitting constant-only model:

  Iteration 0:   log likelihood =  -6408.826  (not concave)
  Iteration 1:   log likelihood = -5808.7085  (not concave)
  Iteration 2:   log likelihood = -5713.8129  (not concave)
  Iteration 3:   log likelihood = -5702.1663
  Iteration 4:   log likelihood = -5679.2492
  Iteration 5:   log likelihood = -5643.1142
  Iteration 6:   log likelihood = -5639.2407
  Iteration 7:   log likelihood = -5638.9706
  Iteration 8:   log likelihood = -5638.9689
  Iteration 9:   log likelihood = -5638.9689

  Fitting full model:

  Iteration 0:   log likelihood = -5638.9689
  Iteration 1:   log likelihood = -5613.8311
  Iteration 2:   log likelihood = -5613.1529
  Iteration 3:   log likelihood =  -5613.144
  Iteration 4:   log likelihood =  -5613.144

  Zero-inflated negative binomial regression            Number of obs =  4,658
  Inflation model: logit                                Nonzero obs   =  1,614
                                                        Zero obs      =  3,044
                                                        LR chi2(5)    =  51.65
  Log likelihood = -5613.144                            Prob > chi2   = 0.0000
```

cv_events	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
cv_events						
sex						
Female	-.0912856	.0852846	-1.07	0.284	-.2584404	.0758691
hypertension						
Yes	.1194127	.0928101	1.29	0.198	-.0624917	.3013171
cholesterol	.0004797	.0009159	0.52	0.600	-.0013153	.0022748
age	.0307398	.005302	5.80	0.000	.020348	.0411316
bmi	.0117543	.0086407	1.36	0.174	-.005181	.0286897
_cons	-10.38916	.3924342	-26.47	0.000	-11.15832	-9.620004
ln(person~e)	1	(exposure)				
inflate						
sex						
Female	1.032707	.1917709	5.39	0.000	.6568428	1.408571
hypertension						
Yes	-1.084939	.3009305	-3.61	0.000	-1.674752	-.4951262
cholesterol	-.0085656	.0022526	-3.80	0.000	-.0129806	-.0041506
age	-.0713819	.012294	-5.81	0.000	-.0954777	-.0472862
_cons	3.905189	.6994889	5.58	0.000	2.534216	5.276162
/lnalpha	.8552979	.082245	10.40	0.000	.6941007	1.016495
alpha	2.352075	.1934464			2.001908	2.763492

```
179 estimates store hypertension_coded

180
181 *now compare all 5 models
182 estimates stats bp_as_is cat_sbp cat_dbp both_cat hypertension_coded
```

Akaike's information criterion and Bayesian information criterion

Model	N	ll(null)	ll(model)	df	AIC	BIC
bp_as_is	4,658	-5634.048	-5606.596	13	11239.19	11323
cat_sbp	4,658	-5639.946	-5609.007	13	11244.01	11327.82
cat_dbp	4,658	-5634.048	-5606.058	13	11238.12	11321.92
both_cat	4,658	-5639.946	-5610.187	13	11246.37	11330.18
hypertensi~d	4,658	-5638.969	-5613.144	12	11250.29	11327.64

Note: BIC uses N = number of observations. See **[R] BIC note**.

```
183
184 *the model with dbp as categorical and sbp as continous had the lowest aic/bic; so i
  > t will be the best going forward
185
186 zinb cv_events i.sex sbp i.high_dbp cholesterol age bmi, inflate(i.sex sbp cholester
  > ol age) exposure(persontime)

  Fitting constant-only model:

  Iteration 0:   log likelihood =  -6408.826  (not concave)
  Iteration 1:   log likelihood = -5828.8948  (not concave)
  Iteration 2:   log likelihood = -5765.1166  (not concave)
  Iteration 3:   log likelihood = -5750.3108  (not concave)
  Iteration 4:   log likelihood = -5742.9957  (not concave)
  Iteration 5:   log likelihood = -5729.1703  (not concave)
  Iteration 6:   log likelihood =  -5722.544  (not concave)
  Iteration 7:   log likelihood = -5648.6204
  Iteration 8:   log likelihood =  -5636.114
  Iteration 9:   log likelihood = -5634.0555
  Iteration 10:  log likelihood = -5634.0476
  Iteration 11:  log likelihood = -5634.0476
```

```
Fitting full model:

Iteration 0:   log likelihood = -5634.0476
Iteration 1:   log likelihood = -5606.8014
Iteration 2:   log likelihood = -5606.0658
Iteration 3:   log likelihood =  -5606.058
Iteration 4:   log likelihood = -5606.0579

Zero-inflated negative binomial regression            Number of obs =  4,658
Inflation model: logit                                Nonzero obs   =  1,614
                                                      Zero obs      =  3,044
                                                      LR chi2(6)    =  55.98
Log likelihood = -5606.058                            Prob > chi2   = 0.0000
```

cv_events	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
cv_events						
sex						
Female	-.0811715	.0855654	-0.95	0.343	-.2488766	.0865335
sbp	.0012645	.0020715	0.61	0.542	-.0027956	.0053245
high_dbp						
Yes	.2554634	.1280647	1.99	0.046	.0044613	.5064656
cholesterol	.0004767	.0009077	0.53	0.599	-.0013024	.0022558
age	.0296087	.0052668	5.62	0.000	.019286	.0399315
bmi	.0067369	.0086987	0.77	0.439	-.0103122	.0237861
_cons	-10.35211	.4060612	-25.49	0.000	-11.14798	-9.556249
ln(person~e)	1	(exposure)				
inflate						
sex						
Female	.9555056	.1819343	5.25	0.000	.5989209	1.31209
sbp	-.0236899	.0048955	-4.84	0.000	-.033285	-.0140948
cholesterol	-.0077326	.0021511	-3.59	0.000	-.0119486	-.0035166
age	-.0658073	.0119001	-5.53	0.000	-.0891311	-.0424836
_cons	6.395834	.8150403	7.85	0.000	4.798384	7.993283
/lnalpha	.8158388	.0845813	9.65	0.000	.6500625	.9816151
alpha	2.261071	.1912443			1.915661	2.668763

```
187
188 *NOW WE DETERMINE IF BMI SHOULD BE CATEGORIZED
189 *MODEL 1;bmi as is(continous)
190 zinb cv_events i.sex sbp i.high_dbp cholesterol age bmi, inflate(i.sex sbp cholester
  > ol age) exposure(persontime)

Fitting constant-only model:

Iteration 0:   log likelihood =  -6408.826  (not concave)
Iteration 1:   log likelihood = -5828.8948  (not concave)
Iteration 2:   log likelihood = -5765.1166  (not concave)
Iteration 3:   log likelihood = -5750.3108  (not concave)
Iteration 4:   log likelihood = -5742.9957  (not concave)
Iteration 5:   log likelihood = -5729.1703  (not concave)
Iteration 6:   log likelihood =  -5722.544  (not concave)
Iteration 7:   log likelihood = -5648.6204
Iteration 8:   log likelihood =  -5636.114
Iteration 9:   log likelihood = -5634.0555
Iteration 10:  log likelihood = -5634.0476
Iteration 11:  log likelihood = -5634.0476

Fitting full model:

Iteration 0:   log likelihood = -5634.0476
Iteration 1:   log likelihood = -5606.8014
Iteration 2:   log likelihood = -5606.0658
Iteration 3:   log likelihood =  -5606.058
Iteration 4:   log likelihood = -5606.0579
```

```
Zero-inflated negative binomial regression              Number of obs =  4,658
Inflation model: logit                                  Nonzero obs   =  1,614
                                                        Zero obs      =  3,044
                                                        LR chi2(6)    =  55.98
Log likelihood = -5606.058                              Prob > chi2   = 0.0000
```

cv_events	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
cv_events						
sex						
Female	-.0811715	.0855654	-0.95	0.343	-.2488766	.0865335
sbp	.0012645	.0020715	0.61	0.542	-.0027956	.0053245
high_dbp						
Yes	.2554634	.1280647	1.99	0.046	.0044613	.5064656
cholesterol	.0004767	.0009077	0.53	0.599	-.0013024	.0022558
age	.0296087	.0052668	5.62	0.000	.019286	.0399315
bmi	.0067369	.0086987	0.77	0.439	-.0103122	.0237861
_cons	-10.35211	.4060612	-25.49	0.000	-11.14798	-9.556249
ln(person~e)	1	(exposure)				
inflate						
sex						
Female	.9555056	.1819343	5.25	0.000	.5989209	1.31209
sbp	-.0236899	.0048955	-4.84	0.000	-.033285	-.0140948
cholesterol	-.0077326	.0021511	-3.59	0.000	-.0119486	-.0035166
age	-.0658073	.0119001	-5.53	0.000	-.0891311	-.0424836
_cons	6.395834	.8150403	7.85	0.000	4.798384	7.993283
/lnalpha	.8158388	.0845813	9.65	0.000	.6500625	.9816151
alpha	2.261071	.1912443			1.915661	2.668763

```
191 estimates store bmi_cont

192
193
194 *MODEL 2;bmi as binary
195 zinb cv_events i.sex sbp i.high_dbp cholesterol age i.obese_ow, inflate(i.sex sbp ch
  > olesterol age) exposure(persontime)

  Fitting constant-only model:

  Iteration 0:   log likelihood =  -6408.826  (not concave)
  Iteration 1:   log likelihood = -5828.8948  (not concave)
  Iteration 2:   log likelihood = -5765.1166  (not concave)
  Iteration 3:   log likelihood = -5750.3108  (not concave)
  Iteration 4:   log likelihood = -5742.9957  (not concave)
  Iteration 5:   log likelihood = -5729.1703  (not concave)
  Iteration 6:   log likelihood =  -5722.544  (not concave)
  Iteration 7:   log likelihood = -5648.6204
  Iteration 8:   log likelihood =  -5636.114
  Iteration 9:   log likelihood = -5634.0555
  Iteration 10:  log likelihood = -5634.0476
  Iteration 11:  log likelihood = -5634.0476

  Fitting full model:

  Iteration 0:   log likelihood = -5634.0476
  Iteration 1:   log likelihood = -5606.4247
  Iteration 2:   log likelihood = -5605.6839
  Iteration 3:   log likelihood = -5605.6761
  Iteration 4:   log likelihood = -5605.6761

  Zero-inflated negative binomial regression              Number of obs =  4,658
  Inflation model: logit                                  Nonzero obs   =  1,614
                                                          Zero obs      =  3,044
                                                          LR chi2(6)    =  56.74
  Log likelihood = -5605.676                              Prob > chi2   = 0.0000
```

cv_events	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
cv_events						
sex						
Female	-.071058	.0859948	-0.83	0.409	-.2396046	.0974887
sbp	.0012766	.0020561	0.62	0.535	-.0027532	.0053065
high_dbp						
Yes	.2564435	.1278895	2.01	0.045	.0057847	.5071023
cholesterol	.0004198	.0009098	0.46	0.644	-.0013634	.0022031
age	.0293583	.0052743	5.57	0.000	.0190208	.0396958
obese_ow						
Yes	.087308	.0746323	1.17	0.242	-.0589687	.2335847
_cons	-10.20829	.3722061	-27.43	0.000	-10.9378	-9.478777
ln(person~e)	1	(exposure)				
inflate						
sex						
Female	.9572429	.1817206	5.27	0.000	.6010771	1.313409
sbp	-.0234569	.0048742	-4.81	0.000	-.0330102	-.0139036
cholesterol	-.0077302	.0021477	-3.60	0.000	-.0119396	-.0035208
age	-.0658287	.0118712	-5.55	0.000	-.0890958	-.0425616
_cons	6.369092	.8112866	7.85	0.000	4.778999	7.959184
/lnalpha	.8131559	.084771	9.59	0.000	.6470079	.979304
alpha	2.255013	.1911597			1.909818	2.662602

```
196 estimates store bmi_cat

197
198
199 *COMPARE THE TWO MODELS;
200 estimates stats bmi_cont bmi_cat
```

Akaike's information criterion and Bayesian information criterion

Model	N	ll(null)	ll(model)	df	AIC	BIC
bmi_cont	4,658	-5634.048	-5606.058	13	11238.12	11321.92
bmi_cat	4,658	-5634.048	-5605.676	13	11237.35	11321.15

Note: BIC uses N = number of observations. See **[R] BIC note**.

```
201
202 *Both AIC/BIC agree and are smaller when bmi categorical
203
204 *SO MODEL IS NOW
205 zinb cv_events i.sex sbp i.high_dbp cholesterol age i.obese_ow, inflate(i.sex sbp ch
  > olesterol age) exposure(persontime)
```

Fitting constant-only model:

```
Iteration 0:   log likelihood =  -6408.826  (not concave)
Iteration 1:   log likelihood = -5828.8948  (not concave)
Iteration 2:   log likelihood = -5765.1166  (not concave)
Iteration 3:   log likelihood = -5750.3108  (not concave)
Iteration 4:   log likelihood = -5742.9957  (not concave)
Iteration 5:   log likelihood = -5729.1703  (not concave)
Iteration 6:   log likelihood =  -5722.544  (not concave)
Iteration 7:   log likelihood = -5648.6204
Iteration 8:   log likelihood =  -5636.114
Iteration 9:   log likelihood = -5634.0555
Iteration 10:  log likelihood = -5634.0476
Iteration 11:  log likelihood = -5634.0476
```

Fitting full model:

```
Iteration 0:   log likelihood = -5634.0476
Iteration 1:   log likelihood = -5606.4247
Iteration 2:   log likelihood = -5605.6839
Iteration 3:   log likelihood = -5605.6761
Iteration 4:   log likelihood = -5605.6761

Zero-inflated negative binomial regression            Number of obs =  4,658
Inflation model: logit                                Nonzero obs   =  1,614
                                                      Zero obs      =  3,044
                                                      LR chi2(6)    =  56.74
Log likelihood = -5605.676                            Prob > chi2   = 0.0000
```

cv_events	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
cv_events						
sex						
Female	-.071058	.0859948	-0.83	0.409	-.2396046	.0974887
sbp	.0012766	.0020561	0.62	0.535	-.0027532	.0053065
high_dbp						
Yes	.2564435	.1278895	2.01	0.045	.0057847	.5071023
cholesterol	.0004198	.0009098	0.46	0.644	-.0013634	.0022031
age	.0293583	.0052743	5.57	0.000	.0190208	.0396958
obese_ow						
Yes	.087308	.0746323	1.17	0.242	-.0589687	.2335847
_cons	-10.20829	.3722061	-27.43	0.000	-10.9378	-9.478777
ln(person~e)	1	(exposure)				
inflate						
sex						
Female	.9572429	.1817206	5.27	0.000	.6010771	1.313409
sbp	-.0234569	.0048742	-4.81	0.000	-.0330102	-.0139036
cholesterol	-.0077302	.0021477	-3.60	0.000	-.0119396	-.0035208
age	-.0658287	.0118712	-5.55	0.000	-.0890958	-.0425616
_cons	6.369092	.8112866	7.85	0.000	4.778999	7.959184
/lnalpha	.8131559	.084771	9.59	0.000	.6470079	.979304
alpha	2.255013	.1911597			1.909818	2.662602

```
206
207
208
209 *Assess for collinearity
210 *model;
211 zinb cv_events i.sex sbp i.high_dbp cholesterol age i.obese_ow, inflate(i.sex sbp ch
  > olesterol age) exposure(persontime)

  Fitting constant-only model:

  Iteration 0:   log likelihood =  -6408.826  (not concave)
  Iteration 1:   log likelihood = -5828.8948  (not concave)
  Iteration 2:   log likelihood = -5765.1166  (not concave)
  Iteration 3:   log likelihood = -5750.3108  (not concave)
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  Iteration 7:   log likelihood = -5648.6204
  Iteration 8:   log likelihood =  -5636.114
  Iteration 9:   log likelihood = -5634.0555
  Iteration 10:  log likelihood = -5634.0476
  Iteration 11:  log likelihood = -5634.0476

  Fitting full model:
```

```
Iteration 0:   log likelihood = -5634.0476
Iteration 1:   log likelihood = -5606.4247
Iteration 2:   log likelihood = -5605.6839
Iteration 3:   log likelihood = -5605.6761
Iteration 4:   log likelihood = -5605.6761

Zero-inflated negative binomial regression            Number of obs =  4,658
Inflation model: logit                                Nonzero obs   =  1,614
                                                      Zero obs      =  3,044
                                                      LR chi2(6)    =  56.74
Log likelihood = -5605.676                            Prob > chi2   = 0.0000
```

cv_events	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
cv_events						
sex						
Female	-.071058	.0859948	-0.83	0.409	-.2396046	.0974887
sbp	.0012766	.0020561	0.62	0.535	-.0027532	.0053065
high_dbp						
Yes	.2564435	.1278895	2.01	0.045	.0057847	.5071023
cholesterol	.0004198	.0009098	0.46	0.644	-.0013634	.0022031
age	.0293583	.0052743	5.57	0.000	.0190208	.0396958
obese_ow						
Yes	.087308	.0746323	1.17	0.242	-.0589687	.2335847
_cons	-10.20829	.3722061	-27.43	0.000	-10.9378	-9.478777
ln(person~e)	1	(exposure)				
inflate						
sex						
Female	.9572429	.1817206	5.27	0.000	.6010771	1.313409
sbp	-.0234569	.0048742	-4.81	0.000	-.0330102	-.0139036
cholesterol	-.0077302	.0021477	-3.60	0.000	-.0119396	-.0035208
age	-.0658287	.0118712	-5.55	0.000	-.0890958	-.0425616
_cons	6.369092	.8112866	7.85	0.000	4.778999	7.959184
/lnalpha	.8131559	.084771	9.59	0.000	.6470079	.979304
alpha	2.255013	.1911597			1.909818	2.662602

```
212
213
214 *predict linear function
215 predict zneglin, xb
  (41 missing values generated)

216
217 *run regress with linear function as outcome
218 regress zneglin i.sex sbp i.high_dbp cholesterol age persontime i.obese_ow
```

Source	SS	df	MS		
Model	1778.87602	7	254.125146	Number of obs	4,658
Residual	369.831316	4,650	.079533616	F(7, 4650)	3195.19
Total	2148.70734	4,657	.46139303	Prob > F	0.0000
				R-squared	0.8279
				Adj R-squared	0.8276
				Root MSE	.28202

zneglin	Coefficient	Std. err.	t	P>|t|	[95% conf.	interval]
sex						
Female	-.0743593	.008545	-8.70	0.000	-.0911116	-.0576071
sbp	.0008594	.0002471	3.48	0.001	.000375	.0013438
high_dbp						
Yes	.2730429	.0169553	16.10	0.000	.2398024	.3062834
cholesterol	.0004262	.0000981	4.34	0.000	.0002339	.0006185
age	.0325905	.0005731	56.86	0.000	.0314669	.0337142
persontime	.0001908	1.29e-06	147.83	0.000	.0001882	.0001933

obese_ow						
Yes	.091143	.00874	10.43	0.000	.0740084	.1082776
_cons	-3.028839	.0415576	-72.88	0.000	-3.110312	-2.947366

```
219
220 *check vif
221 estat vif
```

Variable	VIF	1/VIF
2.sex	1.05	0.950007
sbp	1.86	0.538267
1.high_dbp	1.52	0.655924
cholesterol	1.12	0.894064
age	1.39	0.720815
persontime	1.26	0.794025
1.obese_ow	1.12	0.896440
Mean VIF	1.33	

```
222 **none above 10; collinearity assumption fulfilled;
223 zinb cv_events i.sex sbp i.high_dbp cholesterol age i.obese_ow, inflate(i.sex sbp ch
  > olesterol age) exposure(persontime)

  Fitting constant-only model:

  Iteration 0:   log likelihood =  -6408.826  (not concave)
  Iteration 1:   log likelihood = -5828.8948  (not concave)
  Iteration 2:   log likelihood = -5765.1166  (not concave)
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  Iteration 5:   log likelihood = -5729.1703  (not concave)
  Iteration 6:   log likelihood =  -5722.544  (not concave)
  Iteration 7:   log likelihood = -5648.6204
  Iteration 8:   log likelihood =  -5636.114
  Iteration 9:   log likelihood = -5634.0555
  Iteration 10:  log likelihood = -5634.0476
  Iteration 11:  log likelihood = -5634.0476

  Fitting full model:

  Iteration 0:   log likelihood = -5634.0476
  Iteration 1:   log likelihood = -5606.4247
  Iteration 2:   log likelihood = -5605.6839
  Iteration 3:   log likelihood = -5605.6761
  Iteration 4:   log likelihood = -5605.6761

  Zero-inflated negative binomial regression              Number of obs =  4,658
  Inflation model: logit                                  Nonzero obs   =  1,614
                                                          Zero obs      =  3,044
                                                          LR chi2(6)    =  56.74
  Log likelihood = -5605.676                              Prob > chi2   = 0.0000
```

cv_events	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
cv_events						
sex						
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sbp	.0012766	.0020561	0.62	0.535	-.0027532	.0053065
high_dbp						
Yes	.2564435	.1278895	2.01	0.045	.0057847	.5071023
cholesterol	.0004198	.0009098	0.46	0.644	-.0013634	.0022031
age	.0293583	.0052743	5.57	0.000	.0190208	.0396958
obese_ow						
Yes	.087308	.0746323	1.17	0.242	-.0589687	.2335847
_cons	-10.20829	.3722061	-27.43	0.000	-10.9378	-9.478777
ln(person~e)	1	(exposure)				

inflate						
sex						
Female	**.9572429**	**.1817206**	**5.27**	**0.000**	**.6010771**	**1.313409**
sbp	**-.0234569**	**.0048742**	**-4.81**	**0.000**	**-.0330102**	**-.0139036**
cholesterol	**-.0077302**	**.0021477**	**-3.60**	**0.000**	**-.0119396**	**-.0035208**
age	**-.0658287**	**.0118712**	**-5.55**	**0.000**	**-.0890958**	**-.0425616**
_cons	**6.369092**	**.8112866**	**7.85**	**0.000**	**4.778999**	**7.959184**
/lnalpha	**.8131559**	**.084771**	**9.59**	**0.000**	**.6470079**	**.979304**
alpha	**2.255013**	**.1911597**			**1.909818**	**2.662602**

```
224
225
226
227
228
229 *SELECTION OF VARIABLES TO INCLUDE IN MODEL;
230
231 *CHECK INTERACTION
232 *reduced models
233 zinb cv_events i.sex sbp i.high_dbp cholesterol age i.obese_ow, inflate(i.sex sbp ch
  > olesterol age) exposure(persontime)

  Fitting constant-only model:

  Iteration 0:   log likelihood =  -6408.826  (not concave)
  Iteration 1:   log likelihood = -5828.8948  (not concave)
  Iteration 2:   log likelihood = -5765.1166  (not concave)
  Iteration 3:   log likelihood = -5750.3108  (not concave)
  Iteration 4:   log likelihood = -5742.9957  (not concave)
  Iteration 5:   log likelihood = -5729.1703  (not concave)
  Iteration 6:   log likelihood =  -5722.544  (not concave)
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  Iteration 9:   log likelihood = -5634.0555
  Iteration 10:  log likelihood = -5634.0476
  Iteration 11:  log likelihood = -5634.0476

  Fitting full model:

  Iteration 0:   log likelihood = -5634.0476
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  Iteration 2:   log likelihood = -5605.6839
  Iteration 3:   log likelihood = -5605.6761
  Iteration 4:   log likelihood = -5605.6761

  Zero-inflated negative binomial regression              Number of obs =  4,658
  Inflation model: logit                                  Nonzero obs   =  1,614
                                                          Zero obs      =  3,044
                                                          LR chi2(6)    =  56.74
  Log likelihood = -5605.676                              Prob > chi2   = 0.0000
```

cv_events	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
cv_events						
sex						
Female	**-.071058**	**.0859948**	**-0.83**	**0.409**	**-.2396046**	**.0974887**
sbp	**.0012766**	**.0020561**	**0.62**	**0.535**	**-.0027532**	**.0053065**
high_dbp						
Yes	**.2564435**	**.1278895**	**2.01**	**0.045**	**.0057847**	**.5071023**
cholesterol	**.0004198**	**.0009098**	**0.46**	**0.644**	**-.0013634**	**.0022031**
age	**.0293583**	**.0052743**	**5.57**	**0.000**	**.0190208**	**.0396958**
obese_ow						
Yes	**.087308**	**.0746323**	**1.17**	**0.242**	**-.0589687**	**.2335847**
_cons	**-10.20829**	**.3722061**	**-27.43**	**0.000**	**-10.9378**	**-9.478777**
ln(person~e)	**1**	(exposure)				

inflate						
sex						
Female	**.9572429**	**.1817206**	**5.27**	**0.000**	**.6010771**	**1.313409**
sbp	**-.0234569**	**.0048742**	**-4.81**	**0.000**	**-.0330102**	**-.0139036**
cholesterol	**-.0077302**	**.0021477**	**-3.60**	**0.000**	**-.0119396**	**-.0035208**
age	**-.0658287**	**.0118712**	**-5.55**	**0.000**	**-.0890958**	**-.0425616**
_cons	**6.369092**	**.8112866**	**7.85**	**0.000**	**4.778999**	**7.959184**
/lnalpha	**.8131559**	**.084771**	**9.59**	**0.000**	**.6470079**	**.979304**
alpha	**2.255013**	**.1911597**			**1.909818**	**2.662602**

```
234 estimates store reduced

235
236 *All possible interaction terms included;
237
238 zinb cv_events i.sex sbp i.high_dbp cholesterol age i.obese_ow i.sex#c.sbp i.sex#i.h
  > igh_dbp i.sex#c.cholesterol i.sex#c.age i.sex#i.obese_ow c.sbp#i.high_dbp c.sbp#c.ch
  > olesterol c.sbp#c.age c.sbp#i.obese_ow i.high_dbp#c.cholesterol i.high_dbp#c.age i.h
  > igh_dbp#i.obese_ow c.cholesterol#c.age c.cholesterol#i.obese_ow c.age#i.obese_ow, in
  > flate(i.sex sbp cholesterol age) exposure(persontime)

  Fitting constant-only model:

  Iteration 0:   log likelihood =  -6408.826  (not concave)
  Iteration 1:   log likelihood = -5828.8948  (not concave)
  Iteration 2:   log likelihood = -5765.1166  (not concave)
  Iteration 3:   log likelihood = -5750.3108  (not concave)
  Iteration 4:   log likelihood = -5742.9957  (not concave)
  Iteration 5:   log likelihood = -5729.1703  (not concave)
  Iteration 6:   log likelihood =  -5722.544  (not concave)
  Iteration 7:   log likelihood = -5648.6204
  Iteration 8:   log likelihood =  -5636.114
  Iteration 9:   log likelihood = -5634.0555
  Iteration 10:  log likelihood = -5634.0476
  Iteration 11:  log likelihood = -5634.0476

  Fitting full model:

  Iteration 0:   log likelihood = -5634.0476
  Iteration 1:   log likelihood =  -5588.195
  Iteration 2:   log likelihood = -5585.3806
  Iteration 3:   log likelihood =  -5585.303
  Iteration 4:   log likelihood = -5585.3026

  Zero-inflated negative binomial regression              Number of obs =  4,658
  Inflation model: logit                                  Nonzero obs   =  1,614
                                                          Zero obs      =  3,044
                                                          LR chi2(21)   =  97.49
  Log likelihood = -5585.303                              Prob > chi2   = 0.0000

  ------------------------------------------------------------------------------
    cv_events | Coefficient  Std. err.      z    P>|z|     [95% conf. interva
  > l]
  ------------------------------------------------------------------------------
    cv_events |
          sex |
       Female |   2.486944   .7136332     3.48   0.000     1.088248    3.8856
  > 39
          sbp |  -.0092477   .0165094    -0.56   0.575    -.0416056    .02311
  > 01
              |
    high_dbp |
          Yes |  -.2814333   1.173603    -0.24   0.810    -2.581653    2.0187
  > 86
  cholesterol |   .0146485   .0066888     2.19   0.029     .0015388    .02775
  > 83
```

age	.0370279	.0408046	0.91	0.364	-.0429476	.11700
obese_ow Yes	-.4160764	.65786	-0.63	0.527	-1.705458	.87330
sex#c.sbp Female	-.0121577	.0041138	-2.96	0.003	-.0202205	-.00409
sex#high_dbp Female#Yes	-.1510007	.2633441	-0.57	0.566	-.6671456	.36514
sex#c.cholesterol Female	-.0022684	.0017369	-1.31	0.192	-.0056727	.00113
sex#c.age Female	-.0060191	.0097349	-0.62	0.536	-.0250991	.01306
sex#obese_ow Female#Yes	.0581282	.1520977	0.38	0.702	-.2399779	.35623
high_dbp#c.sbp Yes	.0077974	.0049888	1.56	0.118	-.0019804	.01757
c.sbp#c.cholesterol	-3.25e-06	.0000449	-0.07	0.942	-.0000913	.00008
c.sbp#c.age	.0004003	.0002648	1.51	0.131	-.0001187	.00091
obese_ow#c.sbp Yes	-.0056823	.0040151	-1.42	0.157	-.0135519	.00218
high_dbp#c.cholesterol Yes	-.0000761	.002875	-0.03	0.979	-.005711	.00555
high_dbp#c.age Yes	-.0186031	.0176027	-1.06	0.291	-.0531037	.01589
high_dbp#obese_ow Yes#Yes	.3377095	.2866065	1.18	0.239	-.224029	.89944
c.cholesterol#c.age	-.0002714	.0001034	-2.63	0.009	-.0004741	-.00006
obese_ow#c.cholesterol Yes	.0017902	.0017154	1.04	0.297	-.0015718	.00515
obese_ow#c.age Yes	.0165882	.0094702	1.75	0.080	-.0019732	.03514
_cons	-11.95516	2.199495	-5.44	0.000	-16.26609	-7.6442
ln(persontime)	1	(exposure)				
inflate						
sex Female	1.264707	.2313898	5.47	0.000	.8111915	1.7182

```
> 23
                 sbp |  -.0272683   .0052296    -5.21   0.000    -.0375181   -.01701
> 85
         cholesterol |  -.0064881   .0021582    -3.01   0.003    -.0107182    -.0022
> 58
                 age |  -.0632617   .0118373    -5.34   0.000    -.0864624    -.0400
> 61
               _cons |   6.265896   .8223816     7.62   0.000     4.654057    7.8777
> 34
-------------------------------------------------------------------------------
  ——
            /lnalpha |   .7833005   .0836306     9.37   0.000     .6193876    .94721
> 33
-------------------------------------------------------------------------------
  ——
               alpha |   2.188684   .1830409                      1.85779    2.5785
> 14
-------------------------------------------------------------------------------
  ——

239
240 estimates store full

241
242 *do likelihood test on both models above;
243 lrtest ( reduced) ( full), stats dir

  Likelihood-ratio test
  Assumption: reduced nested within full

  LR chi2(15) =  40.75
  Prob > chi2 = 0.0003

  Akaike's information criterion and Bayesian information criterion

-----------------------------------------------------------------------------
       Model |          N   ll(null)  ll(model)      df        AIC        BIC
-------------+---------------------------------------------------------------
     reduced |      4,658  -5634.048  -5605.676      13   11237.35   11321.15
        full |      4,658  -5634.048  -5585.303      28   11226.61    11407.1
-----------------------------------------------------------------------------
  Note: BIC uses N = number of observations. See [R] BIC note.

-------------------------------------------------------------------------------
        Name | Command    Dependent variable  Number of param.  Title
-------------+-----------------------------------------------------------------
     reduced | zinb       cv_events                       17  Zero-inflated negative
             |                                                binomial regression
        full | zinb       cv_events                       50  Zero-inflated negative
             |                                                binomial regression
-------------------------------------------------------------------------------

244
245 *pvalue of  0.0003 and the full model with interactions has a lower -2logs likelihoo
  > d ratio,means that there is interaction
246
247 *lets drop most non significant term and compare;
248 *drop  high_dbp#c.cholesterol
249 zinb cv_events i.sex sbp i.high_dbp cholesterol age i.obese_ow i.sex#c.sbp i.sex#i.h
  > igh_dbp i.sex#c.cholesterol i.sex#c.age i.sex#i.obese_ow c.sbp#i.high_dbp c.sbp#c.ch
  > olesterol c.sbp#c.age c.sbp#i.obese_ow i.high_dbp#c.age i.high_dbp#i.obese_ow c.chol
  > esterol#c.age c.cholesterol#i.obese_ow c.age#i.obese_ow, inflate(i.sex sbp cholester
  > ol age) exposure(persontime)

  Fitting constant-only model:
```

```
Iteration 0:   log likelihood =  -6408.826  (not concave)
Iteration 1:   log likelihood = -5828.8948  (not concave)
Iteration 2:   log likelihood = -5765.1166  (not concave)
Iteration 3:   log likelihood = -5750.3108  (not concave)
Iteration 4:   log likelihood = -5742.9957  (not concave)
Iteration 5:   log likelihood = -5729.1703  (not concave)
Iteration 6:   log likelihood =  -5722.544  (not concave)
Iteration 7:   log likelihood = -5648.6204
Iteration 8:   log likelihood =  -5636.114
Iteration 9:   log likelihood = -5634.0555
Iteration 10:  log likelihood = -5634.0476
Iteration 11:  log likelihood = -5634.0476

Fitting full model:

Iteration 0:   log likelihood = -5634.0476
Iteration 1:   log likelihood = -5588.1447
Iteration 2:   log likelihood = -5585.3814
Iteration 3:   log likelihood = -5585.3033
Iteration 4:   log likelihood = -5585.3029

Zero-inflated negative binomial regression              Number of obs =  4,658
Inflation model: logit                                  Nonzero obs   =  1,614
                                                        Zero obs      =  3,044
                                                        LR chi2(20)   =  97.49
Log likelihood = -5585.303                              Prob > chi2   = 0.0000
```

cv_events	Coefficient	Std. err.	z	P>|z|	[95% conf. interval]
cv_events					
sex					
Female	**2.487099**	**.713576**	**3.49**	**0.000**	**1.088515**	**3.8856**
sbp	**-.0090885**	**.0153744**	**-0.59**	**0.554**	**-.0392217**	**.02104**
high_dbp					
Yes	**-.2971912**	**1.011478**	**-0.29**	**0.769**	**-2.279652**	**1.6852**
cholesterol	**.0147189**	**.0061396**	**2.40**	**0.017**	**.0026855**	**.02675**
age	**.0369083**	**.0405517**	**0.91**	**0.363**	**-.0425717**	**.11638**
obese_ow					
Yes	**-.4162881**	**.6577616**	**-0.63**	**0.527**	**-1.705477**	**.87290**
sex#c.sbp					
Female	**-.012158**	**.0041136**	**-2.96**	**0.003**	**-.0202205**	**-.00409**
sex#high_dbp					
Female#Yes	**-.1514955**	**.2626735**	**-0.58**	**0.564**	**-.666326**	**.36333**
sex#c.cholesterol					
Female	**-.0022658**	**.0017342**	**-1.31**	**0.191**	**-.0056649**	**.00113**
sex#c.age					
Female	**-.0060318**	**.0097232**	**-0.62**	**0.535**	**-.025089**	**.01302**
sex#obese_ow					
Female#Yes	**.0580681**	**.15208**	**0.38**	**0.703**	**-.2400032**	**.35613**

high_dbp#c.sbp Yes	.007802	.0049856	1.56	0.118	-.0019697	.01757
c.sbp#c.cholesterol	-3.97e-06	.0000358	-0.11	0.912	-.0000741	.00006
c.sbp#c.age	.0004006	.0002647	1.51	0.130	-.0001181	.00091
obese_ow#c.sbp Yes	-.0056773	.0040106	-1.42	0.157	-.0135379	.00218
high_dbp#c.age Yes	-.0186502	.0175131	-1.06	0.287	-.0529753	.01567
high_dbp#obese_ow Yes#Yes	.3364875	.2828616	1.19	0.234	-.217911	.8908
c.cholesterol#c.age	-.0002711	.0001024	-2.65	0.008	-.0004717	-.00007
obese_ow#c.cholesterol Yes	.0017875	.0017123	1.04	0.297	-.0015685	.00514
obese_ow#c.age Yes	.0165929	.0094686	1.75	0.080	-.0019653	.03515
_cons	-11.97059	2.120736	-5.64	0.000	-16.12716	-7.8140
ln(persontime)	1	(exposure)				
inflate						
sex Female	1.264826	.231359	5.47	0.000	.8113706	1.7182
sbp	-.0272645	.0052271	-5.22	0.000	-.0375094	-.01701
cholesterol	-.0064888	.0021579	-3.01	0.003	-.0107183	-.00225
age	-.0632647	.0118366	-5.34	0.000	-.086464	-.04006
_cons	6.26566	.8222623	7.62	0.000	4.654055	7.8772
/lnalpha	.7832935	.0836274	9.37	0.000	.6193867	.94720
alpha	2.188669	.1830327			1.857788	2.578

```
250 estimates store minushigh_dbpcholesterol

251
252 lrtest (minushigh_dbpcholesterol) ( full), stats dir

  Likelihood-ratio test
  Assumption: minushigh_db~l nested within full

   LR chi2(1) =    0.00
  Prob > chi2 = 0.9789

  Akaike's information criterion and Bayesian information criterion

┌──────────────┬──────────────────────────────────────────────────────────────┐
│        Model │          N   ll(null)  ll(model)      df        AIC        BIC │
├──────────────┼──────────────────────────────────────────────────────────────┤
│ minushigh_~l │      4,658  -5634.048  -5585.303      27   11224.61   11398.66 │
│         full │      4,658  -5634.048  -5585.303      28   11226.61    11407.1 │
└──────────────┴──────────────────────────────────────────────────────────────┘
  Note: BIC uses N = number of observations. See [R] BIC note.

┌──────────────┬──────────────────────────────────────────────────────────────┐
│         Name │ Command    Dependent variable     Number of param.  Title    │
├──────────────┼──────────────────────────────────────────────────────────────┤
│ minushigh_~l │ zinb       cv_events                        48  Zero-inflated negative binomial regression │
│         full │ zinb       cv_events                        50  Zero-inflated negative binomial regression │
└──────────────┴──────────────────────────────────────────────────────────────┘

253 *but looking at the log likelihood of the two models; dropping high_dbp#c.cholestero
  > l does not change the log likelhood; so we shall drop high_dbp#c.cholesterol even LR
  >  test is signficant
254
255 *significant p value means we keep high_dbpcholesterol
256
257 *drop  sbp#cholesterol
258
259 zinb cv_events i.sex sbp i.high_dbp cholesterol age i.obese_ow i.sex#c.sbp i.sex#i.h
  > igh_dbp i.sex#c.cholesterol i.sex#c.age i.sex#i.obese_ow c.sbp#i.high_dbp c.sbp#c.ag
  > e c.sbp#i.obese_ow i.high_dbp#c.age i.high_dbp#i.obese_ow c.cholesterol#c.age c.chol
  > esterol#i.obese_ow c.age#i.obese_ow, inflate(i.sex sbp cholesterol age) exposure(per
  > sontime)

  Fitting constant-only model:

  Iteration 0:   log likelihood =  -6408.826  (not concave)
  Iteration 1:   log likelihood = -5828.8948  (not concave)
  Iteration 2:   log likelihood = -5765.1166  (not concave)
  Iteration 3:   log likelihood = -5750.3108  (not concave)
  Iteration 4:   log likelihood = -5742.9957  (not concave)
  Iteration 5:   log likelihood = -5729.1703  (not concave)
  Iteration 6:   log likelihood =  -5722.544  (not concave)
  Iteration 7:   log likelihood = -5648.6204
  Iteration 8:   log likelihood =  -5636.114
  Iteration 9:   log likelihood = -5634.0555
  Iteration 10:  log likelihood = -5634.0476
  Iteration 11:  log likelihood = -5634.0476

  Fitting full model:

  Iteration 0:   log likelihood = -5634.0476
  Iteration 1:   log likelihood = -5588.1492
  Iteration 2:   log likelihood = -5585.3877
  Iteration 3:   log likelihood = -5585.3095
  Iteration 4:   log likelihood = -5585.3091
```

Zero-inflated negative binomial regression — Number of obs = **4,658**
Inflation model: **logit** — Nonzero obs = **1,614**
Zero obs = **3,044**
LR chi2(**19**) = **97.48**
Log likelihood = **-5585.309** — Prob > chi2 = **0.0000**

cv_events	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
cv_events						
sex						
Female	**2.496082**	**.7094402**	**3.52**	**0.000**	**1.105605**	**3.8865**
sbp	**-.0098501**	**.0137568**	**-0.72**	**0.474**	**-.0368129**	**.01711**
high_dbp						
Yes	**-.2941102**	**1.011102**	**-0.29**	**0.771**	**-2.275834**	**1.6876**
cholesterol	**.0143184**	**.0049671**	**2.88**	**0.004**	**.0045831**	**.02405**
age	**.0378364**	**.0397066**	**0.95**	**0.341**	**-.0399871**	**.11565**
obese_ow						
Yes	**-.4045279**	**.6494918**	**-0.62**	**0.533**	**-1.677508**	**.86845**
sex#c.sbp						
Female	**-.0122035**	**.0040941**	**-2.98**	**0.003**	**-.0202277**	**-.00417**
sex#high_dbp						
Female#Yes	**-.1527772**	**.2624229**	**-0.58**	**0.560**	**-.6671166**	**.36156**
sex#c.cholesterol						
Female	**-.0023**	**.0017071**	**-1.35**	**0.178**	**-.005646**	**.00104**
sex#c.age						
Female	**-.0059391**	**.0096869**	**-0.61**	**0.540**	**-.024925**	**.01304**
sex#obese_ow						
Female#Yes	**.0597153**	**.1513689**	**0.39**	**0.693**	**-.2369622**	**.35639**
high_dbp#c.sbp						
Yes	**.0078035**	**.0049866**	**1.56**	**0.118**	**-.0019702**	**.01757**
c.sbp#c.age	**.0003978**	**.0002634**	**1.51**	**0.131**	**-.0001185**	**.00091**
obese_ow#c.sbp						
Yes	**-.0056996**	**.0040063**	**-1.42**	**0.155**	**-.0135518**	**.00215**
high_dbp#c.age						
Yes	**-.0186721**	**.0175117**	**-1.07**	**0.286**	**-.0529945**	**.01565**
high_dbp#obese_ow						
Yes#Yes	**.3344837**	**.2822833**	**1.18**	**0.236**	**-.2187813**	**.88774**
c.cholesterol#c.age	**-.0002735**	**.0001**	**-2.73**	**0.006**	**-.0004695**	**-.00007**

obese_ow#c.cholesterol Yes	.00174	.0016583	1.05	0.294	-.0015102	.00499
obese_ow#c.age Yes	.0166272	.009464	1.76	0.079	-.0019219	.03517
_cons	-11.89832	2.019154	-5.89	0.000	-15.85579	-7.9408
ln(persontime)	1	(exposure)				
inflate						
sex Female	1.264999	.2314041	5.47	0.000	.8114556	1.7185
sbp	-.0272908	.0052214	-5.23	0.000	-.0375246	-.01705
cholesterol	-.0065319	.0021249	-3.07	0.002	-.0106966	-.00236
age	-.0632037	.0118237	-5.35	0.000	-.0863777	-.04002
_cons	6.275345	.8181746	7.67	0.000	4.671752	7.8789
/lnalpha	.7833907	.0835873	9.37	0.000	.6195626	.94721
alpha	2.188882	.1829628			1.858115	2.5785

```
260 estimates store minussbpcholesterol

261
262 lrtest (minushigh_dbpcholesterol) (minussbpcholesterol), stats dir
```

Likelihood-ratio test
Assumption: minussbpchol~l nested within minushigh_db~l

LR chi2(**1**) = **0.01**
Prob > chi2 = **0.9116**

Akaike's information criterion and Bayesian information criterion

Model	N	ll(null)	ll(model)	df	AIC	BIC
minussbpch~l	**4,658**	**-5634.048**	**-5585.309**	**26**	**11222.62**	**11390.22**
minushigh_~l	**4,658**	**-5634.048**	**-5585.303**	**27**	**11224.61**	**11398.66**

Note: BIC uses N = number of observations. See **[R] BIC note**.

Name	Command	Dependent variable	Number of param.	Title
minussbpch~l	**zinb**	**cv_events**	**47**	***Zero-inflated negative binomial regression***
minushigh_~l	**zinb**	**cv_events**	**48**	***Zero-inflated negative binomial regression***

```
263
264 *significant p value means we keep sbpcholestorol; However looking the at the -2xLL
  > of the the two models; the model without the sbpcholestorol; has the smaller -2xLL;
  > therefore ; we drop it
265
266
267 *drop  high_dbp
268 zinb cv_events i.sex sbp cholesterol age i.obese_ow i.sex#c.sbp i.sex#i.high_dbp i.s
  > ex#c.cholesterol i.sex#c.age i.sex#i.obese_ow c.sbp#i.high_dbp c.sbp#c.age c.sbp#i.o
  > bese_ow i.high_dbp#c.age i.high_dbp#i.obese_ow c.cholesterol#c.age c.cholesterol#i.o
  > bese_ow c.age#i.obese_ow, inflate(i.sex sbp cholesterol age) exposure(persontime)
  note: 1.high_dbp#1.obese_ow omitted because of collinearity.

  Fitting constant-only model:

  Iteration 0:   log likelihood =  -6408.826  (not concave)
  Iteration 1:   log likelihood = -5828.8948  (not concave)
  Iteration 2:   log likelihood = -5765.1166  (not concave)
  Iteration 3:   log likelihood = -5750.3108  (not concave)
  Iteration 4:   log likelihood = -5742.9957  (not concave)
  Iteration 5:   log likelihood = -5729.1703  (not concave)
  Iteration 6:   log likelihood =  -5722.544  (not concave)
  Iteration 7:   log likelihood = -5648.6204
  Iteration 8:   log likelihood =  -5636.114
  Iteration 9:   log likelihood = -5634.0555
  Iteration 10:  log likelihood = -5634.0476
  Iteration 11:  log likelihood = -5634.0476

  Fitting full model:

  Iteration 0:   log likelihood = -5634.0476
  Iteration 1:   log likelihood = -5588.1492
  Iteration 2:   log likelihood = -5585.3877
  Iteration 3:   log likelihood = -5585.3095
  Iteration 4:   log likelihood = -5585.3091

  Zero-inflated negative binomial regression              Number of obs =  4,658
  Inflation model: logit                                  Nonzero obs   =  1,614
                                                          Zero obs      =  3,044
                                                          LR chi2(19)   =  97.48
  Log likelihood = -5585.309                              Prob > chi2   = 0.0000

  ------------------------------------------------------------------------------
    __
               cv_events | Coefficient  Std. err.      z    P>|z|     [95% conf. interva
  > l]
  ------------------------------------------------------------------------------
    __
  cv_events              |
                     sex |
                  Female |   2.496082   .7094402     3.52   0.000     1.105605    3.8865
  > 59
                     sbp |  -.0098501   .0137568    -0.72   0.474    -.0368129     .01711
  > 27
             cholesterol |   .0143184   .0049671     2.88   0.004     .0045831    .02405
  > 37
                     age |   .0378364   .0397066     0.95   0.341    -.0399871    .11565
  > 99
                         |
                obese_ow |
                     Yes |  -.4045279   .6494918    -0.62   0.533    -1.677508    .86845
  > 26
                         |
                sex#c.sbp |
                  Female |  -.0122035   .0040941    -2.98   0.003    -.0202277   -.00417
  > 93
                         |
            sex#high_dbp |
               Male#Yes |   .0403735   .9591429     0.04   0.966    -1.839512    1.9202
  > 59
             Female#Yes |  -.1124036    1.02745    -0.11   0.913    -2.126168    1.9013
  > 61
```

sex#c.cholesterol Female	-.0023	.0017071	-1.35	0.178	-.005646	.00104
sex#c.age Female	-.0059391	.0096869	-0.61	0.540	-.024925	.01304
sex#obese_ow Female#Yes	.0597153	.1513689	0.39	0.693	-.2369622	.35639
high_dbp#c.sbp Yes	.0078035	.0049866	1.56	0.118	-.0019702	.01757
c.sbp#c.age	.0003978	.0002634	1.51	0.131	-.0001185	.00091
obese_ow#c.sbp Yes	-.0056996	.0040063	-1.42	0.155	-.0135518	.00215
high_dbp#c.age Yes	-.0186721	.0175117	-1.07	0.286	-.0529945	.01565
high_dbp#obese_ow Yes#No	-.3344837	.2822833	-1.18	0.236	-.8877488	.21878
Yes#Yes	0	(omitted)				
c.cholesterol#c.age	-.0002735	.0001	-2.73	0.006	-.0004695	-.00007
obese_ow#c.cholesterol Yes	.00174	.0016583	1.05	0.294	-.0015102	.00499
obese_ow#c.age Yes	.0166272	.009464	1.76	0.079	-.0019219	.03517
_cons	-11.89832	2.019154	-5.89	0.000	-15.85579	-7.9408
ln(persontime)	1	(exposure)				
inflate						
sex Female	1.264999	.2314041	5.47	0.000	.8114556	1.7185
sbp	-.0272908	.0052214	-5.23	0.000	-.0375246	-.01705
cholesterol	-.0065319	.0021249	-3.07	0.002	-.0106966	-.00236
age	-.0632037	.0118237	-5.35	0.000	-.0863777	-.04002
_cons	6.275345	.8181746	7.67	0.000	4.671752	7.8789
/lnalpha	.7833907	.0835873	9.37	0.000	.6195626	.94721
alpha	2.188882	.1829628			1.858115	2.5785

```
269 estimates store minushigh_dbp

270
271 *lrtest (minushigh_dbp) (minussbpcholesterol), stats dir
272 *models cant be compared becauses; "df(unrestricted) = df(restricted) = 26"
273 *however looking at the -2 log likelihood for both models; they look exactly the sam
  > e; so droping  high_dbp wont weaken our model
274
275 *verdict; drop high_dbp
276
277
278 *drop sex#high_dbp
279 zinb cv_events i.sex sbp cholesterol age i.obese_ow i.sex#c.sbp i.sex#c.cholesterol
  > i.sex#c.age i.sex#i.obese_ow c.sbp#i.high_dbp c.sbp#c.age c.sbp#i.obese_ow i.high_db
  > p#c.age i.high_dbp#i.obese_ow c.cholesterol#c.age c.cholesterol#i.obese_ow c.age#i.o
  > bese_ow, inflate(i.sex sbp cholesterol age) exposure(persontime)

  Fitting constant-only model:

  Iteration 0:   log likelihood =  -6408.826  (not concave)
  Iteration 1:   log likelihood = -5828.8948  (not concave)
  Iteration 2:   log likelihood = -5765.1166  (not concave)
  Iteration 3:   log likelihood = -5750.3108  (not concave)
  Iteration 4:   log likelihood = -5742.9957  (not concave)
  Iteration 5:   log likelihood = -5729.1703  (not concave)
  Iteration 6:   log likelihood =  -5722.544  (not concave)
  Iteration 7:   log likelihood = -5648.6204
  Iteration 8:   log likelihood =  -5636.114
  Iteration 9:   log likelihood = -5634.0555
  Iteration 10:  log likelihood = -5634.0476
  Iteration 11:  log likelihood = -5634.0476

  Fitting full model:

  Iteration 0:   log likelihood = -5634.0476
  Iteration 1:   log likelihood = -5588.2876
  Iteration 2:   log likelihood = -5585.5579
  Iteration 3:   log likelihood = -5585.4785
  Iteration 4:   log likelihood = -5585.4781

  Zero-inflated negative binomial regression            Number of obs =  4,658
  Inflation model: logit                                Nonzero obs   =  1,614
                                                        Zero obs      =  3,044
                                                        LR chi2(18)   =  97.14
  Log likelihood = -5585.478                            Prob > chi2   = 0.0000

  ------------------------------------------------------------------------------
    cv_events | Coefficient  Std. err.      z    P>|z|     [95% conf. interva
  > l]
  ------------------------------------------------------------------------------
    cv_events |
          sex |
       Female |   2.639856   .6658081     3.96   0.000     1.334896    3.9448
  > 16
          sbp |  -.0101127   .0137472    -0.74   0.462    -.0370566    .01683
  > 12
  cholesterol |   .0144442   .0049625     2.91   0.004     .0047178    .02417
  > 06
          age |   .0356151   .0395382     0.90   0.368    -.0418783    .11310
  > 86
              |
    obese_ow |
          Yes |  -.3811679    .648672    -0.59   0.557    -1.652542    .89020
  > 59
              |
    sex#c.sbp |
       Female |  -.0135754    .003348    -4.05   0.000    -.0201375   -.00701
  > 34
              |
  sex#c.cholesterol |
```

Female	-.002283	.0017073	-1.34	0.181	-.0056293	.00106
sex#c.age Female	-.0055601	.0096716	-0.57	0.565	-.0245161	.01339
sex#obese_ow Female#Yes	.0597182	.1514893	0.39	0.693	-.2371955	.35663
high_dbp#c.sbp Yes	.0072407	.0048887	1.48	0.139	-.0023409	.01682
c.sbp#c.age	.0004213	.0002605	1.62	0.106	-.0000892	.00093
obese_ow#c.sbp Yes	-.005797	.0040061	-1.45	0.148	-.0136488	.00205
high_dbp#c.age Yes	-.0198216	.0174315	-1.14	0.255	-.0539868	.01434
high_dbp#obese_ow Yes#No	-.2271157	1.005491	-0.23	0.821	-2.197842	1.743
Yes#Yes	.1166391	.951213	0.12	0.902	-1.747704	1.9809
c.cholesterol#c.age	-.0002761	.0000999	-2.76	0.006	-.0004719	-.00008
obese_ow#c.cholesterol Yes	.0017209	.0016585	1.04	0.299	-.0015297	.00497
obese_ow#c.age Yes	.0164714	.0094639	1.74	0.082	-.0020775	.03502
_cons	-11.89926	2.01889	-5.89	0.000	-15.85621	-7.9423
ln(persontime)	1	(exposure)				
inflate						
sex Female	1.267486	.2329273	5.44	0.000	.8109571	1.7240
sbp	-.0273593	.0052448	-5.22	0.000	-.037639	-.01707
cholesterol	-.0065293	.0021314	-3.06	0.002	-.0107068	-.00235
age	-.0632039	.011877	-5.32	0.000	-.0864825	-.03992
_cons	6.275509	.8220523	7.63	0.000	4.664316	7.8867
/lnalpha	.7864249	.0833936	9.43	0.000	.6229763	.94987
alpha	2.195533	.1830935			1.864469	2.5853

```
280 estimates store minussexhigh_dbp

281
282
283 lrtest (minushigh_dbp) (minussexhigh_dbp), stats dir
```

Likelihood-ratio test
Assumption: minussexhigh~p nested within minushigh_dbp

LR chi2(**1**) = **0.34**
Prob > chi2 = **0.5610**

Akaike's information criterion and Bayesian information criterion

Model	N	ll(null)	ll(model)	df	AIC	BIC
minussexhi~p	**4,658**	**-5634.048**	**-5585.478**	**25**	**11220.96**	**11382.11**
minushigh_~p	**4,658**	**-5634.048**	**-5585.309**	**26**	**11222.62**	**11390.22**

Note: BIC uses N = number of observations. See **[R] BIC note**.

Name	Command	Dependent variable	Number of param.	Title
minussexhi~p	**zinb**	**cv_events**	**41**	***Zero-inflated negative binomial regression***
minushigh_~p	**zinb**	**cv_events**	**45**	***Zero-inflated negative binomial regression***

```
284 *nonsignificant LR test; drop sex#high_dbp
285
286
287
288
289 *drop  high_dbpobese_ow
290 zinb cv_events i.sex sbp cholesterol age i.obese_ow i.sex#c.sbp i.sex#c.cholesterol
  > i.sex#c.age i.sex#i.obese_ow c.sbp#i.high_dbp c.sbp#c.age c.sbp#i.obese_ow i.high_db
  > p#c.age c.cholesterol#c.age c.cholesterol#i.obese_ow c.age#i.obese_ow, inflate(i.sex
  >  sbp cholesterol age) exposure(persontime)

Fitting constant-only model:

Iteration 0:   log likelihood =  -6408.826  (not concave)
Iteration 1:   log likelihood = -5828.8948  (not concave)
Iteration 2:   log likelihood = -5765.1166  (not concave)
Iteration 3:   log likelihood = -5750.3108  (not concave)
Iteration 4:   log likelihood = -5742.9957  (not concave)
Iteration 5:   log likelihood = -5729.1703  (not concave)
Iteration 6:   log likelihood =  -5722.544  (not concave)
Iteration 7:   log likelihood = -5648.6204
Iteration 8:   log likelihood =  -5636.114
Iteration 9:   log likelihood = -5634.0555
Iteration 10:  log likelihood = -5634.0476
Iteration 11:  log likelihood = -5634.0476

Fitting full model:

Iteration 0:   log likelihood = -5634.0476
Iteration 1:   log likelihood = -5588.9018
Iteration 2:   log likelihood = -5586.3092
Iteration 3:   log likelihood = -5586.2327
Iteration 4:   log likelihood = -5586.2323

Zero-inflated negative binomial regression            Number of obs =  4,658
Inflation model: logit                                Nonzero obs   =  1,614
                                                      Zero obs      =  3,044
                                                      LR chi2(16)   =  95.63
Log likelihood = -5586.232                            Prob > chi2   = 0.0000
```

cv_events	Coefficient	Std. err.	z	P>|z|	[95% conf. interval]
cv_events					
sex					
Female	**2.587287**	**.6621596**	**3.91**	**0.000**	**1.289478**	**3.8850**
sbp	**-.0102785**	**.0126601**	**-0.81**	**0.417**	**-.0350917**	**.01453**
cholesterol	**.0145628**	**.0049615**	**2.94**	**0.003**	**.0048384**	**.02428**
age	**.0389616**	**.0371154**	**1.05**	**0.294**	**-.0337832**	**.11170**
obese_ow					
Yes	**-.6658793**	**.6021607**	**-1.11**	**0.269**	**-1.846093**	**.51433**
sex#c.sbp					
Female	**-.0134001**	**.0033388**	**-4.01**	**0.000**	**-.0199441**	**-.00685**
sex#c.cholesterol					
Female	**-.0022138**	**.0017027**	**-1.30**	**0.194**	**-.0055511**	**.00112**
sex#c.age					
Female	**-.0052009**	**.0096531**	**-0.54**	**0.590**	**-.0241206**	**.01371**
sex#obese_ow					
Female#Yes	**.048678**	**.1511692**	**0.32**	**0.747**	**-.2476082**	**.34496**
high_dbp#c.sbp					
Yes	**.0075465**	**.0042017**	**1.80**	**0.072**	**-.0006887**	**.01578**
c.sbp#c.age	**.000391**	**.0002358**	**1.66**	**0.097**	**-.0000712**	**.00085**
obese_ow#c.sbp					
Yes	**-.0032279**	**.0033898**	**-0.95**	**0.341**	**-.0098718**	**.0034**
high_dbp#c.age					
Yes	**-.0205412**	**.0137332**	**-1.50**	**0.135**	**-.0474578**	**.00637**
c.cholesterol#c.age	**-.0002761**	**.0000999**	**-2.76**	**0.006**	**-.0004718**	**-.00008**
obese_ow#c.cholesterol					
Yes	**.0015982**	**.0016547**	**0.97**	**0.334**	**-.001645**	**.00484**
obese_ow#c.age					
Yes	**.0166138**	**.0094685**	**1.75**	**0.079**	**-.0019442**	**.03517**
_cons	**-11.893**	**1.906079**	**-6.24**	**0.000**	**-15.62885**	**-8.1571**
ln(persontime)	**1**	(exposure)			
inflate					
sex					
Female	**1.259594**	**.2312853**	**5.45**	**0.000**	**.8062832**	**1.7129**

```
                 sbp |  -.0272958   .0052759    -5.17   0.000    -.0376363   -.01695
> 53
         cholesterol |   -.006406   .0021258    -3.01   0.003    -.0105725   -.00223
> 95
                 age |  -.0634131   .0118441    -5.35   0.000    -.0866272    -.0401
> 99
               _cons |   6.257632   .8231972     7.60   0.000     4.644195    7.8710
> 68
-------------------------------------------------------------------------------
  ——
            /lnalpha |   .7858124   .0836432     9.39   0.000     .6218747    .94975
> 02
-------------------------------------------------------------------------------
  ——
               alpha |   2.194189   .1835291                      1.862416    2.5850
> 64
-------------------------------------------------------------------------------
  ——
```

```
291 estimates store minushigh_dbpobese_ow

292
293 lrtest (minushigh_dbpobese_ow) (minussexhigh_dbp), stats dir

  Likelihood-ratio test
  Assumption: minushigh_db~w nested within minussexhigh~p

   LR chi2(2) =   1.51
  Prob > chi2 = 0.4704

  Akaike's information criterion and Bayesian information criterion

-----------------------------------------------------------------------------
       Model |          N   ll(null)  ll(model)      df        AIC        BIC
-------------+---------------------------------------------------------------
minushigh_~w |      4,658  -5634.048  -5586.232      23   11218.46   11366.73
minussexhi~p |      4,658  -5634.048  -5585.478      25   11220.96   11382.11
-----------------------------------------------------------------------------
  Note: BIC uses N = number of observations. See [R] BIC note.

-------------------------------------------------------------------------------
        Name | Command    Dependent variable    Number of param.  Title
-------------+-----------------------------------------------------------------
minushigh_~w | zinb       cv_events                       37   Zero-inflated negative
             |                                                 binomial regression
minussexhi~p | zinb       cv_events                       41   Zero-inflated negative
             |                                                 binomial regression
-------------------------------------------------------------------------------

294 *nonsignificant LR; DROP high_dbpobese_ow
295
296
297 *drop sex#obese_ow
298 zinb cv_events i.sex sbp cholesterol age i.obese_ow i.sex#c.sbp i.sex#c.cholesterol
  > i.sex#c.age c.sbp#i.high_dbp c.sbp#c.age c.sbp#i.obese_ow i.high_dbp#c.age c.cholest
  > erol#c.age c.cholesterol#i.obese_ow c.age#i.obese_ow, inflate(i.sex sbp cholesterol
  > age) exposure(persontime)

  Fitting constant-only model:
```

```
Iteration 0:   log likelihood =  -6408.826  (not concave)
Iteration 1:   log likelihood = -5828.8948  (not concave)
Iteration 2:   log likelihood = -5765.1166  (not concave)
Iteration 3:   log likelihood = -5750.3108  (not concave)
Iteration 4:   log likelihood = -5742.9957  (not concave)
Iteration 5:   log likelihood = -5729.1703  (not concave)
Iteration 6:   log likelihood =  -5722.544  (not concave)
Iteration 7:   log likelihood = -5648.6204
Iteration 8:   log likelihood =  -5636.114
Iteration 9:   log likelihood = -5634.0555
Iteration 10:  log likelihood = -5634.0476
Iteration 11:  log likelihood = -5634.0476

Fitting full model:

Iteration 0:   log likelihood = -5634.0476
Iteration 1:   log likelihood =  -5588.953
Iteration 2:   log likelihood = -5586.3622
Iteration 3:   log likelihood = -5586.2846
Iteration 4:   log likelihood = -5586.2842

Zero-inflated negative binomial regression              Number of obs =  4,658
Inflation model: logit                                  Nonzero obs   =  1,614
                                                        Zero obs      =  3,044
                                                        LR chi2(15)   =  95.53
Log likelihood = -5586.284                              Prob > chi2   = 0.0000
```

cv_events	Coefficient	Std. err.	z	P>|z|	[95% conf. interval]
cv_events					
sex					
Female	2.571358	.6599208	3.90	0.000	1.277937 3.8647
sbp	-.0101095	.0126513	-0.80	0.424	-.0349057 .01468
cholesterol	.0145152	.0049596	2.93	0.003	.0047945 .02423
age	.0398408	.0370233	1.08	0.282	-.0327236 .11240
obese_ow					
Yes	-.6855904	.5986861	-1.15	0.252	-1.858994 .48781
sex#c.sbp					
Female	-.0131943	.0032755	-4.03	0.000	-.0196142 -.00677
sex#c.cholesterol					
Female	-.0021645	.001695	-1.28	0.202	-.0054867 .00115
sex#c.age					
Female	-.0051007	.0096436	-0.53	0.597	-.0240018 .01380
high_dbp#c.sbp					
Yes	.0074414	.00419	1.78	0.076	-.0007708 .01565
c.sbp#c.age	.0003845	.0002349	1.64	0.102	-.0000759 .00084
obese_ow#c.sbp					
Yes	-.0030858	.0033599	-0.92	0.358	-.0096712 .00349
high_dbp#c.age					

```
                    Yes |  -.0202021   .0136967    -1.47   0.140    -.0470471     .0066
> 43
                        |
   c.cholesterol#c.age |  -.0002766   .0000999    -2.77   0.006    -.0004723   -.00008
> 09
                        |
obese_ow#c.cholesterol |
                    Yes |   .0016647    .001641     1.01   0.310    -.0015515     .0048
> 81
                        |
        obese_ow#c.age |
                    Yes |   .0167927   .0094477     1.78   0.075    -.0017244     .03530
> 98
                        |
                  _cons | -11.91519   1.905437    -6.25   0.000    -15.64978   -8.1806
> 05
       ln(persontime) |          1  (exposure)
----------------------+-------------------------------------------------------------
  --
inflate
                        |
                    sex |
                 Female |   1.262801   .2317816     5.45   0.000     .8085172    1.7170
> 84
                    sbp |  -.0273049   .0052806    -5.17   0.000    -.0376547    -.0169
> 55
            cholesterol |  -.0064097   .0021276    -3.01   0.003    -.0105797   -.00223
> 97
                    age |  -.0634784   .0118562    -5.35   0.000    -.0867161   -.04024
> 06
                  _cons |   6.258687   .8239084     7.60   0.000     4.643856    7.8735
> 18
----------------------+-------------------------------------------------------------
  --
               /lnalpha |   .7867231   .0835386     9.42   0.000     .6229904     .95045
> 58
----------------------+-------------------------------------------------------------
  --
                  alpha |   2.196188   .1834665                      1.864495    2.5868
> 88
----------------------+-------------------------------------------------------------
  --
```

```
299 estimates store minussexobese_ow

300
301
302 lrtest (minushigh_dbpobese_ow) (minussexobese_ow), stats dir

  Likelihood-ratio test
  Assumption: minussexobes~w nested within minushigh_db~w

   LR chi2(1) =    0.10
  Prob > chi2 = 0.7474

  Akaike's information criterion and Bayesian information criterion

-------------------------------------------------------------------------------
       Model |          N   ll(null)  ll(model)      df        AIC        BIC
-------------+-----------------------------------------------------------------
minussexob~w |      4,658  -5634.048  -5586.284      22   11216.57   11358.39
minushigh_~w |      4,658  -5634.048  -5586.232      23   11218.46   11366.73
-------------------------------------------------------------------------------
  Note: BIC uses N = number of observations. See [R] BIC note.
```

Name	Command	Dependent variable	Number of param.	Title
minussexob~w	**zinb**	**cv_events**	**33**	***Zero-inflated negative binomial regression***
minushigh_~w	**zinb**	**cv_events**	**37**	***Zero-inflated negative binomial regression***

```
303 *nonsignificant; drop sex#obese_ow
304
305
306 *drop sex#c.age
307 zinb cv_events i.sex sbp cholesterol age i.obese_ow i.sex#c.sbp i.sex#c.cholesterol
  > c.sbp#i.high_dbp c.sbp#c.age c.sbp#i.obese_ow i.high_dbp#c.age c.cholesterol#c.age c
  > .cholesterol#i.obese_ow c.age#i.obese_ow, inflate(i.sex sbp cholesterol age) exposur
  > e(persontime)

  Fitting constant-only model:

  Iteration 0:   log likelihood =  -6408.826  (not concave)
  Iteration 1:   log likelihood = -5828.8948  (not concave)
  Iteration 2:   log likelihood = -5765.1166  (not concave)
  Iteration 3:   log likelihood = -5750.3108  (not concave)
  Iteration 4:   log likelihood = -5742.9957  (not concave)
  Iteration 5:   log likelihood = -5729.1703  (not concave)
  Iteration 6:   log likelihood =  -5722.544  (not concave)
  Iteration 7:   log likelihood = -5648.6204
  Iteration 8:   log likelihood =  -5636.114
  Iteration 9:   log likelihood = -5634.0555
  Iteration 10:  log likelihood = -5634.0476
  Iteration 11:  log likelihood = -5634.0476

  Fitting full model:

  Iteration 0:   log likelihood = -5634.0476
  Iteration 1:   log likelihood = -5588.8991
  Iteration 2:   log likelihood = -5586.4977
  Iteration 3:   log likelihood = -5586.4245
  Iteration 4:   log likelihood = -5586.4242

  Zero-inflated negative binomial regression              Number of obs =  4,658
  Inflation model: logit                                  Nonzero obs   =  1,614
                                                          Zero obs      =  3,044
                                                          LR chi2(14)   =  95.25
  Log likelihood = -5586.424                              Prob > chi2   = 0.0000

  ------------------------------------------------------------------------------
    cv_events | Coefficient  Std. err.      z    P>|z|     [95% conf. interva
  > l]
  ------------------------------------------------------------------------------
    cv_events |
          sex |
       Female |   2.403402   .5775536     4.16   0.000     1.271418    3.5353
  > 86
          sbp |  -.0099396   .0126417    -0.79   0.432     -.034717    .01483
  > 77
  cholesterol |   .0145858    .004963     2.94   0.003     .0048584    .02431
  > 31
          age |   .0383385   .0368893     1.04   0.299    -.0339633    .11064
  > 02
          |
    obese_ow |
          Yes |  -.7028585   .5977187    -1.18   0.240    -1.874366    .46864
  > 86
          |
      sex#c.sbp |
       Female |  -.0135957   .0031832    -4.27   0.000    -.0198345   -.00735
  > 68
```

```
                                  |
      sex#c.cholesterol           |
                Female            |  -.0022759   .0016796    -1.35   0.175    -.0055679    .00101
> 62
                                  |
        high_dbp#c.sbp            |
                  Yes             |   .0075888   .0041805     1.82   0.069    -.0006048    .01578
> 24
                                  |
             c.sbp#c.age          |   .0003816   .0002346     1.63   0.104    -.0000783    .00084
> 14
                                  |
        obese_ow#c.sbp            |
                  Yes             |   -.003124   .0033585    -0.93   0.352    -.0097064    .00345
> 85
                                  |
        high_dbp#c.age            |
                  Yes             |  -.0205383   .0136826    -1.50   0.133    -.0473556     .0062
> 79
                                  |
    c.cholesterol#c.age           |  -.0002786   .0000998    -2.79   0.005    -.0004741    -.0000
> 83
                                  |
obese_ow#c.cholesterol            |
                  Yes             |   .0016434   .0016405     1.00   0.316     -.001572    .00485
> 87
                                  |
        obese_ow#c.age            |
                  Yes             |   .0173244   .0094002     1.84   0.065    -.0010998    .03574
> 85
                                  |
                _cons             | -11.83814   1.899519    -6.23   0.000    -15.56113   -8.1151
> 53
        ln(persontime)            |          1  (exposure)
--------------------------------- +-------------------------------------------------------------
  __
inflate                           |
                                  |
                 sex              |
               Female             |   1.235859   .2214046     5.58   0.000     .8019138    1.6698
> 04
                  sbp             |  -.0272951   .0052659    -5.18   0.000    -.0376161   -.01697
> 41
          cholesterol             |  -.0064445   .0021204    -3.04   0.002    -.0106004   -.00228
> 86
                  age             |   -.062676   .0116894    -5.36   0.000    -.0855868   -.03976
> 53
                _cons             |   6.256804   .8216665     7.61   0.000     4.646367     7.867
> 24
--------------------------------- +-------------------------------------------------------------
  __
             /lnalpha             |   .7832742   .0836914     9.36   0.000     .6192421    .94730
> 63
--------------------------------- +-------------------------------------------------------------
  __
                alpha             |   2.188627   .1831692                      1.85752    2.5787
> 54
--------------------------------- +-------------------------------------------------------------
  __
```

```
308 estimates store minusexage
```

```
309
310
311 lrtest (minusexage) (minussexobese_ow), stats dir
```

Likelihood-ratio test
Assumption: minusexage nested within minussexobes~w

LR chi2(**1**) = **0.28**
Prob > chi2 = **0.5966**

Akaike's information criterion and Bayesian information criterion

Model	N	ll(null)	ll(model)	df	AIC	BIC
minusexage	**4,658**	**-5634.048**	**-5586.424**	**21**	**11214.85**	**11350.22**
minussexob~w	**4,658**	**-5634.048**	**-5586.284**	**22**	**11216.57**	**11358.39**

Note: BIC uses N = number of observations. See **[R] BIC note**.

Name	Command	Dependent variable	Number of param.	Title
minusexage	**zinb**	**cv_events**	**31**	***Zero-inflated negative binomial regression***
minussexob~w	**zinb**	**cv_events**	**33**	***Zero-inflated negative binomial regression***

```
312 *nonsignificant; drop
313
314 *drop sbp
315 zinb cv_events i.sex cholesterol age i.obese_ow i.sex#c.sbp i.sex#c.cholesterol c.sb
  > p#i.high_dbp c.sbp#c.age c.sbp#i.obese_ow i.high_dbp#c.age c.cholesterol#c.age c.cho
  > lesterol#i.obese_ow c.age#i.obese_ow, inflate(i.sex cholesterol age) exposure(person
  > time)
  note: 1.high_dbp#c.sbp omitted because of collinearity.
  note: 1.obese_ow#c.sbp omitted because of collinearity.

  Fitting constant-only model:

  Iteration 0:   log likelihood =  -6408.826  (not concave)
  Iteration 1:   log likelihood = -5832.2281  (not concave)
  Iteration 2:   log likelihood = -5771.4327  (not concave)
  Iteration 3:   log likelihood = -5757.8771
  Iteration 4:   log likelihood = -5713.0225
  Iteration 5:   log likelihood =  -5665.999
  Iteration 6:   log likelihood = -5659.0027
  Iteration 7:   log likelihood =  -5658.389
  Iteration 8:   log likelihood = -5658.3875
  Iteration 9:   log likelihood = -5658.3875

  Fitting full model:

  Iteration 0:   log likelihood = -5658.3875
  Iteration 1:   log likelihood = -5607.1293
  Iteration 2:   log likelihood = -5604.2892
  Iteration 3:   log likelihood = -5604.2192
  Iteration 4:   log likelihood = -5604.2188

  Zero-inflated negative binomial regression      Number of obs =  4,658
  Inflation model: logit                          Nonzero obs   =  1,614
                                                  Zero obs      =  3,044
                                                  LR chi2(14)   = 108.34
  Log likelihood = -5604.219                      Prob > chi2   = 0.0000
```

cv_events	Coefficient	Std. err.	z	P>|z|	[95% conf. interval]
cv_events					
sex					
Female	2.252615	.5725016	3.93	0.000	1.130532 3.37497
cholesterol	.0121053	.0050366	2.40	0.016	.0022337 .02196
age	.0793143	.0360805	2.20	0.028	.0085978 .15008
obese_ow					
Yes	-.6661552	.6027517	-1.11	0.269	-1.847527 .51564
sex#c.sbp					
Male	.0185798	.0114426	1.62	0.104	-.0038472 .04100
Female	.0072258	.0117171	0.62	0.537	-.0157393 .03019
sex#c.cholesterol					
Female	-.0029702	.001712	-1.73	0.083	-.0063257 .00038
high_dbp#c.sbp					
No	-.0059263	.0044231	-1.34	0.180	-.0145955 .00274
Yes	0	(omitted)			
c.sbp#c.age	-.0000505	.000224	-0.23	0.822	-.0004895 .00038
obese_ow#c.sbp					
No	.0043372	.0034695	1.25	0.211	-.0024629 .01113
Yes	0	(omitted)			
high_dbp#c.age					
Yes	-.0155069	.0144372	-1.07	0.283	-.0438034 .01278
c.cholesterol#c.age	-.0002296	.0001012	-2.27	0.023	-.000428 -.00003
obese_ow#c.cholesterol					
Yes	.0014989	.0016447	0.91	0.362	-.0017246 .00472
obese_ow#c.age					
Yes	.0206835	.0094462	2.19	0.029	.0021694 .03919
_cons	-14.62324	1.827564	-8.00	0.000	-18.2052 -11.041
ln(persontime)	1	(exposure)			
inflate					
sex					
Female	1.381444	.253153	5.46	0.000	.8852729 1.8776
cholesterol	-.0094158	.0024555	-3.83	0.000	-.0142285 -.00460
age	-.0903001	.012384	-7.29	0.000	-.1145723 -.06602
_cons	4.34709	.7144429	6.08	0.000	2.946808 5.7473

```
------------------------------+------------------------------------------------------------------------------
  --
             /lnalpha |   .8557722    .081481    10.50   0.000     .6960723    1.0154
> 72
-----------------------------+------------------------------------------------------------------------------
  --
               alpha |   2.353191   .1917403                      2.005859    2.7606
> 66
-----------------------------+------------------------------------------------------------------------------
  --

316 estimates store sbpdropped

317
318 *lrtest (minusexage) (sbpdropped), stats dir
319 *-2LL lowered significantly after dropping sbp
320
321
322 *drop  sex#c.sbp
323 zinb cv_events i.sex cholesterol age i.obese_ow i.sex#c.cholesterol c.sbp#i.high_dbp
  >  c.sbp#c.age c.sbp#i.obese_ow i.high_dbp#c.age c.cholesterol#c.age c.cholesterol#i.o
  > bese_ow c.age#i.obese_ow, inflate(i.sex cholesterol age) exposure(persontime)
  note: 1.obese_ow#c.sbp omitted because of collinearity.

  Fitting constant-only model:

  Iteration 0:   log likelihood =  -6408.826  (not concave)
  Iteration 1:   log likelihood = -5832.2281  (not concave)
  Iteration 2:   log likelihood = -5771.4327  (not concave)
  Iteration 3:   log likelihood = -5757.8771
  Iteration 4:   log likelihood = -5713.0225
  Iteration 5:   log likelihood =  -5665.999
  Iteration 6:   log likelihood = -5659.0027
  Iteration 7:   log likelihood =  -5658.389
  Iteration 8:   log likelihood = -5658.3875
  Iteration 9:   log likelihood = -5658.3875

  Fitting full model:

  Iteration 0:   log likelihood = -5658.3875
  Iteration 1:   log likelihood = -5612.7852
  Iteration 2:   log likelihood = -5610.6822
  Iteration 3:   log likelihood = -5610.6348
  Iteration 4:   log likelihood = -5610.6346

  Zero-inflated negative binomial regression              Number of obs =  4,658
  Inflation model: logit                                  Nonzero obs   =  1,614
                                                          Zero obs      =  3,044
                                                          LR chi2(13)   =  95.51
  Log likelihood = -5610.635                              Prob > chi2   = 0.0000

------------------------------+------------------------------------------------------------------------------
  --
             cv_events | Coefficient  Std. err.      z    P>|z|     [95% conf. interva
> l]
-----------------------------+------------------------------------------------------------------------------
  --
cv_events                    |
                      sex    |
                   Female |   .8393298   .4169095     2.01   0.044     .0222022    1.6564
> 58
              cholesterol |   .0121967   .0050693     2.41   0.016     .0022609    .02213
> 24
                      age |   .0923804   .0360939     2.56   0.010     .0216377    .16312
> 31
                          |
                obese_ow  |
                      Yes |  -.8719859   .6035065    -1.44   0.148    -2.054837    .31086
> 52
                          |
       sex#c.cholesterol  |
                   Female |  -.0036085   .0017075    -2.11   0.035    -.0069552   -.00026
```

```
> 18
                                                                                
        high_dbp#c.sbp |
                    No |   .0119417   .0120853     0.99   0.323     -.011745     .03562
> 84
                   Yes |   .0162692    .011468     1.42   0.156    -.0062078     .03874
> 61
                       |
          c.sbp#c.age |  -.0001574   .0002227    -0.71   0.480    -.0005938      .0002
> 79
                       |
        obese_ow#c.sbp |
                    No |   .0029084   .0034668     0.84   0.402    -.0038864     .00970
> 31
                   Yes |          0  (omitted)
                       |
        high_dbp#c.age |
                   Yes |  -.0095376   .0144456    -0.66   0.509    -.0378506     .01877
> 53
                       |
   c.cholesterol#c.age |  -.0002278   .0001017    -2.24   0.025    -.0004271    -.00002
> 84
                       |
obese_ow#c.cholesterol |
                   Yes |   .0015202   .0016504     0.92   0.357    -.0017146      .0047
> 55
                       |
        obese_ow#c.age |
                   Yes |   .0206992   .0095078     2.18   0.029     .0020642     .03933
> 41
                       |
                 _cons |  -14.30327   1.843015    -7.76   0.000    -17.91551   -10.691
> 03
       ln(persontime) |          1  (exposure)
----------------------+-----------------------------------------------------------
  __
inflate                |
                       |
                   sex |
                Female |   1.277615   .2433374     5.25   0.000     .8006829    1.7545
> 48
           cholesterol |  -.0098413   .0025162    -3.91   0.000    -.0147731    -.00490
> 96
                   age |  -.0891908   .0125916    -7.08   0.000    -.1138698    -.06451
> 18
                 _cons |   4.435077   .7279609     6.09   0.000     3.008299    5.8618
> 54
----------------------+-----------------------------------------------------------
  __
              /lnalpha |   .8760806   .0807478    10.85   0.000     .7178179    1.0343
> 43
----------------------+-----------------------------------------------------------
  __
                 alpha |   2.401469   .1939133                      2.049955    2.8132
> 58
----------------------+-----------------------------------------------------------
  __

324 estimates store minussexsbp
325
```

```
326
327
328 lrtest (minussexsbp) (sbpdropped), stats dir

  Likelihood-ratio test
  Assumption: minussexsbp nested within sbpdropped

   LR chi2(1) =  12.83
  Prob > chi2 = 0.0003

  Akaike's information criterion and Bayesian information criterion

-----------------------------------------------------------------------------
       Model |          N   ll(null)  ll(model)      df        AIC        BIC
-------------+---------------------------------------------------------------
  minussexsbp |      4,658  -5658.388  -5610.635      19   11259.27   11381.75
    sbpdropped |      4,658  -5658.388  -5604.219      20   11248.44   11377.36
-----------------------------------------------------------------------------
  Note: BIC uses N = number of observations. See [R] BIC note.

-------------------------------------------------------------------------------
             |           Dependent  Number of
        Name | Command    variable     param.  Title
-------------+-----------------------------------------------------------------
  minussexsbp | zinb       cv_events        27  Zero-inflated negative
             |                                  binomial regression
    sbpdropped | zinb       cv_events        29  Zero-inflated negative
             |                                  binomial regression
-------------------------------------------------------------------------------

329 *signficant; keep ; also model with sexsbp has lower -2LL
330
331
332 *drop obese_ow#csbp
333 zinb cv_events i.sex cholesterol age i.obese_ow i.sex#c.sbp i.sex#c.cholesterol c.sb
  > p#i.high_dbp c.sbp#c.age i.high_dbp#c.age c.cholesterol#c.age c.cholesterol#i.obese_
  > ow c.age#i.obese_ow, inflate(i.sex cholesterol age) exposure(persontime)
  note: 1.high_dbp#c.sbp omitted because of collinearity.

  Fitting constant-only model:

  Iteration 0:   log likelihood =  -6408.826  (not concave)
  Iteration 1:   log likelihood = -5832.2281  (not concave)
  Iteration 2:   log likelihood = -5771.4327  (not concave)
  Iteration 3:   log likelihood = -5757.8771
  Iteration 4:   log likelihood = -5713.0225
  Iteration 5:   log likelihood =  -5665.999
  Iteration 6:   log likelihood = -5659.0027
  Iteration 7:   log likelihood =  -5658.389
  Iteration 8:   log likelihood = -5658.3875
  Iteration 9:   log likelihood = -5658.3875

  Fitting full model:

  Iteration 0:   log likelihood = -5658.3875
  Iteration 1:   log likelihood =  -5607.839
  Iteration 2:   log likelihood = -5605.0725
  Iteration 3:   log likelihood = -5605.0033
  Iteration 4:   log likelihood = -5605.0029

  Zero-inflated negative binomial regression            Number of obs =  4,658
  Inflation model: logit                                Nonzero obs   =  1,614
                                                        Zero obs      =  3,044
                                                        LR chi2(13)   = 106.77
  Log likelihood = -5605.003                            Prob > chi2   = 0.0000
```

cv_events	Coefficient	Std. err.	z	P>|z|	[95% conf. interval]
cv_events					
sex					
Female	2.178458	.5711465	3.81	0.000	1.059031 3.29785
cholesterol	.0122536	.0050386	2.43	0.015	.0023783 .02212
age	.0855196	.035818	2.39	0.017	.0153177 .15572
obese_ow					
Yes	-1.030219	.5269568	-1.96	0.051	-2.063036 .00259
sex#c.sbp					
Male	.0202636	.0113621	1.78	0.075	-.0020057 .04253
Female	.0093399	.0115849	0.81	0.420	-.0133661 .03204
sex#c.cholesterol					
Female	-.0029191	.0017135	-1.70	0.088	-.0062775 .00043
high_dbp#c.sbp					
No	-.0045005	.0042944	-1.05	0.295	-.0129173 .00391
Yes	0	(omitted)			
c.sbp#c.age	-.0000757	.0002234	-0.34	0.735	-.0005135 .00036
high_dbp#c.age					
Yes	-.0113182	.0141152	-0.80	0.423	-.0389836 .01634
c.cholesterol#c.age	-.0002323	.0001013	-2.29	0.022	-.0004309 -.00003
obese_ow#c.cholesterol					
Yes	.0013636	.0016425	0.83	0.406	-.0018556 .00458
obese_ow#c.age					
Yes	.0167757	.0088984	1.89	0.059	-.0006649 .03421
_cons	-14.60017	1.830843	-7.97	0.000	-18.18856 -11.011
ln(persontime)	1	(exposure)			
inflate					
sex					
Female	1.380585	.254398	5.43	0.000	.8819739 1.8791
cholesterol	-.009467	.0024669	-3.84	0.000	-.014302 -.0046
age	-.0901827	.0124336	-7.25	0.000	-.1145522 -.06581
_cons	4.346417	.7179702	6.05	0.000	2.939221 5.7536
/lnalpha	.8600537	.081176	10.59	0.000	.7009516 1.0191

```
  ——
               alpha |   2.363288   .1918422                      2.01567    2.7708
> 54
  ——
```

```
334 estimates store minusobese_owcsbp

335
336 lrtest (sbpdropped) (minusobese_owcsbp), stats dir
```

Likelihood-ratio test
Assumption: minusobese_o~p nested within sbpdropped

LR chi2(**1**) = **1.57**
Prob > chi2 = **0.2105**

Akaike's information criterion and Bayesian information criterion

Model	N	ll(null)	ll(model)	df	AIC	BIC
minusobese~p	**4,658**	**-5658.388**	**-5605.003**	**19**	**11248.01**	**11370.49**
sbpdropped	**4,658**	**-5658.388**	**-5604.219**	**20**	**11248.44**	**11377.36**

Note: BIC uses N = number of observations. See **[R] BIC note**.

Name	Command	Dependent variable	Number of param.	Title
minusobese~p	**zinb**	**cv_events**	**27**	***Zero-inflated negative binomial regression***
sbpdropped	**zinb**	**cv_events**	**29**	***Zero-inflated negative binomial regression***

```
337 *nonsignificant drop
338
339
340
341 *drop obese_ow#c.cholesterol
342 zinb cv_events i.sex cholesterol age i.obese_ow i.sex#c.sbp i.sex#c.cholesterol c.sb
  > p#i.high_dbp c.sbp#c.age i.high_dbp#c.age c.cholesterol#c.age c.age#i.obese_ow, infl
  > ate(i.sex cholesterol age) exposure(persontime)
  note: 1.high_dbp#c.sbp omitted because of collinearity.

  Fitting constant-only model:

  Iteration 0:   log likelihood =  -6408.826  (not concave)
  Iteration 1:   log likelihood = -5832.2281  (not concave)
  Iteration 2:   log likelihood = -5771.4327  (not concave)
  Iteration 3:   log likelihood = -5757.8771
  Iteration 4:   log likelihood = -5713.0225
  Iteration 5:   log likelihood =  -5665.999
  Iteration 6:   log likelihood = -5659.0027
  Iteration 7:   log likelihood =  -5658.389
  Iteration 8:   log likelihood = -5658.3875
  Iteration 9:   log likelihood = -5658.3875

  Fitting full model:

  Iteration 0:   log likelihood = -5658.3875
  Iteration 1:   log likelihood = -5608.1571
  Iteration 2:   log likelihood = -5605.4162
  Iteration 3:   log likelihood = -5605.3478
  Iteration 4:   log likelihood = -5605.3474
```

Zero-inflated negative binomial regression — Number of obs = **4,658**
Inflation model: **logit** — Nonzero obs = **1,614**
Zero obs = **3,044**
LR chi2(**12**) = **106.08**
Log likelihood = **-5605.347** — Prob > chi2 = **0.0000**

cv_events	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
cv_events						
sex						
Female	**2.225619**	**.567525**	**3.92**	**0.000**	**1.11329**	**3.337948**
cholesterol	**.0129156**	**.0049611**	**2.60**	**0.009**	**.0031919**	**.0226393**
age	**.0829932**	**.0356335**	**2.33**	**0.020**	**.0131528**	**.1528336**
obese_ow						
Yes	**-.7765644**	**.4299684**	**-1.81**	**0.071**	**-1.619287**	**.0661582**
sex#c.sbp						
Male	**.0201863**	**.0113705**	**1.78**	**0.076**	**-.0020994**	**.042472**
Female	**.0092194**	**.0115942**	**0.80**	**0.427**	**-.0135048**	**.0319436**
sex#c.cholesterol						
Female	**-.0030991**	**.0016986**	**-1.82**	**0.068**	**-.0064283**	**.0002301**
high_dbp#c.sbp						
No	**-.0045811**	**.004297**	**-1.07**	**0.286**	**-.0130031**	**.0038408**
Yes	**0**	(omitted)				
c.sbp#c.age	**-.0000727**	**.0002235**	**-0.33**	**0.745**	**-.0005107**	**.0003653**
high_dbp#c.age						
Yes	**-.0114433**	**.0141276**	**-0.81**	**0.418**	**-.039133**	**.0162464**
c.cholesterol#c.age	**-.0002272**	**.0001009**	**-2.25**	**0.024**	**-.0004249**	**-.0000295**
obese_ow#c.age						
Yes	**.0181363**	**.0087547**	**2.07**	**0.038**	**.0009773**	**.0352953**
_cons	**-14.6874**	**1.824763**	**-8.05**	**0.000**	**-18.26387**	**-11.11093**
ln(persontime)	**1**	(exposure)				
inflate						
sex						
Female	**1.384517**	**.2551297**	**5.43**	**0.000**	**.8844714**	**1.884562**
cholesterol	**-.0093859**	**.0024559**	**-3.82**	**0.000**	**-.0141994**	**-.0045723**
age	**-.0902365**	**.0124375**	**-7.26**	**0.000**	**-.1146135**	**-.0658595**
_cons	**4.327079**	**.717121**	**6.03**	**0.000**	**2.921548**	**5.732611**
/lnalpha	**.8610866**	**.0811016**	**10.62**	**0.000**	**.7021303**	**1.020043**
alpha	**2.36573**	**.1918645**			**2.018047**	**2.773314**

```
343 estimates store minusobesechole

344
345
346 lrtest (minusobesechole) (minusobese_owcsbp), stats dir

  Likelihood-ratio test
  Assumption: minusobesech~e nested within minusobese_o~p

   LR chi2(1) =    0.69
  Prob > chi2 = 0.4065

  Akaike's information criterion and Bayesian information criterion
```

Model	N	ll(null)	ll(model)	df	AIC	BIC
minusobese~e	**4,658**	**-5658.388**	**-5605.347**	**18**	**11246.69**	**11362.73**
minusobese~p	**4,658**	**-5658.388**	**-5605.003**	**19**	**11248.01**	**11370.49**

Note: BIC uses N = number of observations. See **[R] BIC note**.

Name	Command	Dependent variable	Number of param.	Title
minusobese~e	**zinb**	**cv_events**	**25**	***Zero-inflated negative binomial regression***
minusobese~p	**zinb**	**cv_events**	**27**	***Zero-inflated negative binomial regression***

```
347 *nonsignificant drop obese_ow#c.cholesterol
348
349 *drop c.sbp#c.age
350 zinb cv_events i.sex cholesterol age i.obese_ow i.sex#c.sbp i.sex#c.cholesterol c.sb
  > p#i.high_dbp i.high_dbp#c.age c.cholesterol#c.age c.age#i.obese_ow, inflate(i.sex ch
  > olesterol age) exposure(persontime)
  note: 1.high_dbp#c.sbp omitted because of collinearity.

  Fitting constant-only model:

  Iteration 0:   log likelihood =  -6408.826  (not concave)
  Iteration 1:   log likelihood = -5832.2281  (not concave)
  Iteration 2:   log likelihood = -5771.4327  (not concave)
  Iteration 3:   log likelihood = -5757.8771
  Iteration 4:   log likelihood = -5713.0225
  Iteration 5:   log likelihood =  -5665.999
  Iteration 6:   log likelihood = -5659.0027
  Iteration 7:   log likelihood =  -5658.389
  Iteration 8:   log likelihood = -5658.3875
  Iteration 9:   log likelihood = -5658.3875

  Fitting full model:

  Iteration 0:   log likelihood = -5658.3875
  Iteration 1:   log likelihood = -5607.9631
  Iteration 2:   log likelihood = -5605.4741
  Iteration 3:   log likelihood = -5605.4007
  Iteration 4:   log likelihood = -5605.4002

  Zero-inflated negative binomial regression            Number of obs =  4,658
  Inflation model: logit                                Nonzero obs   =  1,614
                                                        Zero obs      =  3,044
                                                        LR chi2(11)   = 105.97
  Log likelihood = -5605.4                              Prob > chi2   = 0.0000
```

cv_events	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
cv_events						
sex						
Female	**2.235047**	**.5674963**	**3.94**	**0.000**	**1.122774**	**3.347319**
cholesterol	**.0130959**	**.0049353**	**2.65**	**0.008**	**.0034229**	**.0227688**
age	**.0744527**	**.0241197**	**3.09**	**0.002**	**.0271791**	**.1217264**
obese_ow						
Yes	**-.7485014**	**.4214728**	**-1.78**	**0.076**	**-1.574573**	**.07757**
sex#c.sbp						
Male	**.0167989**	**.0045535**	**3.69**	**0.000**	**.0078742**	**.0257236**
Female	**.0056882**	**.0040575**	**1.40**	**0.161**	**-.0022643**	**.0136407**
sex#c.cholesterol						
Female	**-.0030712**	**.0016959**	**-1.81**	**0.070**	**-.0063951**	**.0002527**

high_dbp#c.sbp						
No	**-.0049154**	**.0041653**	**-1.18**	**0.238**	**-.0130792**	**.0032484**
Yes	**0**	(omitted)				
high_dbp#c.age						
Yes	**-.0125968**	**.0136519**	**-0.92**	**0.356**	**-.0393541**	**.0141604**
c.cholesterol#c.age	**-.0002308**	**.0001003**	**-2.30**	**0.021**	**-.0004275**	**-.0000342**
obese_ow#c.age						
Yes	**.0176034**	**.0086042**	**2.05**	**0.041**	**.0007395**	**.0344673**
_cons	**-14.25091**	**1.238267**	**-11.51**	**0.000**	**-16.67787**	**-11.82396**
ln(persontime)	**1**	(exposure)				
inflate						
sex						
Female	**1.38129**	**.2538435**	**5.44**	**0.000**	**.8837664**	**1.878815**
cholesterol	**-.0093675**	**.0024514**	**-3.82**	**0.000**	**-.0141721**	**-.004563**
age	**-.0907463**	**.0123654**	**-7.34**	**0.000**	**-.114982**	**-.0665106**
_cons	**4.349475**	**.7131733**	**6.10**	**0.000**	**2.951681**	**5.747269**
/lnalpha	**.8605728**	**.0810585**	**10.62**	**0.000**	**.7017011**	**1.019445**
alpha	**2.364515**	**.191664**			**2.017181**	**2.771655**

```
351 estimates store minussbpage

352
353 lrtest (minusobesechole) (minussbpage), stats dir

  Likelihood-ratio test
  Assumption: minussbpage nested within minusobesech~e

   LR chi2(1) =   0.11
  Prob > chi2 = 0.7451

  Akaike's information criterion and Bayesian information criterion
```

Model	N	ll(null)	ll(model)	df	AIC	BIC
minussbpage	**4,658**	**-5658.388**	**-5605.4**	**17**	**11244.8**	**11354.39**
minusobese~e	**4,658**	**-5658.388**	**-5605.347**	**18**	**11246.69**	**11362.73**

Note: BIC uses N = number of observations. See **[R] BIC note**.

Name	Command	Dependent variable	Number of param.	Title
minussbpage	**zinb**	**cv_events**	**24**	***Zero-inflated negative binomial regression***
minusobese~e	**zinb**	**cv_events**	**25**	***Zero-inflated negative binomial regression***

```
354 *nonsignificant drop sbpage
355
```

```
356
357 *drop high_dbp#c.age
358 zinb cv_events i.sex cholesterol age i.obese_ow i.sex#c.sbp i.sex#c.cholesterol c.sb
  > p#i.high_dbp c.cholesterol#c.age c.age#i.obese_ow, inflate(i.sex cholesterol age) ex
  > posure(persontime)
  note: 1.high_dbp#c.sbp omitted because of collinearity.

  Fitting constant-only model:

  Iteration 0:   log likelihood =  -6408.826  (not concave)
  Iteration 1:   log likelihood = -5832.2281  (not concave)
  Iteration 2:   log likelihood = -5771.4327  (not concave)
  Iteration 3:   log likelihood = -5757.8771
  Iteration 4:   log likelihood = -5713.0225
  Iteration 5:   log likelihood =  -5665.999
  Iteration 6:   log likelihood = -5659.0027
  Iteration 7:   log likelihood =  -5658.389
  Iteration 8:   log likelihood = -5658.3875
  Iteration 9:   log likelihood = -5658.3875

  Fitting full model:

  Iteration 0:   log likelihood = -5658.3875
  Iteration 1:   log likelihood = -5608.3406
  Iteration 2:   log likelihood = -5605.8989
  Iteration 3:   log likelihood = -5605.8267
  Iteration 4:   log likelihood = -5605.8263

  Zero-inflated negative binomial regression              Number of obs =  4,658
  Inflation model: logit                                  Nonzero obs   =  1,614
                                                          Zero obs      =  3,044
                                                          LR chi2(10)   = 105.12
  Log likelihood = -5605.826                              Prob > chi2   = 0.0000
```

cv_events	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
cv_events						
sex						
Female	2.212843	.5652936	3.91	0.000	1.104888	3.320798
cholesterol	.013288	.0049237	2.70	0.007	.0036377	.0229383
age	.0744792	.024098	3.09	0.002	.0272479	.1217104
obese_ow						
Yes	-.7277217	.4209954	-1.73	0.084	-1.552858	.097414
sex#c.sbp						
Male	.0133398	.0025515	5.23	0.000	.0083389	.0183407
Female	.0024734	.0020545	1.20	0.229	-.0015533	.0065
sex#c.cholesterol						
Female	-.0030931	.0016945	-1.83	0.068	-.0064142	.000228
high_dbp#c.sbp						
No	-.0011499	.0007844	-1.47	0.143	-.0026873	.0003875
Yes	0	(omitted)				
c.cholesterol#c.age	-.0002349	.0001001	-2.35	0.019	-.0004312	-.0000387
obese_ow#c.age						
Yes	.0172301	.0085978	2.00	0.045	.0003788	.0340815
_cons	-14.29498	1.235723	-11.57	0.000	-16.71695	-11.873
ln(persontime)	1	(exposure)				
inflate						
sex						
Female	1.381769	.2534404	5.45	0.000	.8850346	1.878503
cholesterol	-.0093395	.0024461	-3.82	0.000	-.0141339	-.0045452
age	-.0906595	.0123653	-7.33	0.000	-.114895	-.066424
_cons	4.342371	.7124278	6.10	0.000	2.946038	5.738704

/lnalpha	**.8589691**	**.0812086**	**10.58**	**0.000**	**.6998031**	**1.018135**
alpha	**2.360726**	**.1917113**			**2.013356**	**2.768028**

```
359 estimates store minusdbpage

360
361 lrtest (minusdbpage) (minussbpage), stats dir

  Likelihood-ratio test
  Assumption: minusdbpage nested within minussbpage

   LR chi2(1) =   0.85
  Prob > chi2 = 0.3560

  Akaike's information criterion and Bayesian information criterion
```

Model	N	ll(null)	ll(model)	df	AIC	BIC
minusdbpage	**4,658**	**-5658.388**	**-5605.826**	**16**	**11243.65**	**11346.79**
minussbpage	**4,658**	**-5658.388**	**-5605.4**	**17**	**11244.8**	**11354.39**

Note: BIC uses N = number of observations. See **[R] BIC note**.

Name	Command	Dependent variable	Number of param.	Title
minusdbpage	**zinb**	**cv_events**	**22**	***Zero-inflated negative binomial regression***
minussbpage	**zinb**	**cv_events**	**24**	***Zero-inflated negative binomial regression***

```
362 *nonsignificant drop high_dbp#c.age
363
364 * drop sex#c.sbp
365 zinb cv_events i.sex cholesterol age i.obese_ow i.sex#c.cholesterol c.sbp#i.high_dbp
  >  c.cholesterol#c.age c.age#i.obese_ow, inflate(i.sex cholesterol age) exposure(perso
  > ntime)

  Fitting constant-only model:

  Iteration 0:   log likelihood =  -6408.826  (not concave)
  Iteration 1:   log likelihood = -5832.2281  (not concave)
  Iteration 2:   log likelihood = -5771.4327  (not concave)
  Iteration 3:   log likelihood = -5757.8771
  Iteration 4:   log likelihood = -5713.0225
  Iteration 5:   log likelihood =  -5665.999
  Iteration 6:   log likelihood = -5659.0027
  Iteration 7:   log likelihood =  -5658.389
  Iteration 8:   log likelihood = -5658.3875
  Iteration 9:   log likelihood = -5658.3875

  Fitting full model:

  Iteration 0:   log likelihood = -5658.3875
  Iteration 1:   log likelihood = -5613.7696
  Iteration 2:   log likelihood = -5611.9434
  Iteration 3:   log likelihood = -5611.8949
  Iteration 4:   log likelihood = -5611.8947

  Zero-inflated negative binomial regression            Number of obs =  4,658
  Inflation model: logit                                Nonzero obs   =  1,614
                                                        Zero obs      =  3,044
                                                        LR chi2(9)    =  92.99
  Log likelihood = -5611.895                            Prob > chi2   = 0.0000
```

cv_events	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
cv_events						
sex						
Female	**.8548607**	**.4108657**	**2.08**	**0.037**	**.0495787**	**1.660143**
cholesterol	**.0135734**	**.0049604**	**2.74**	**0.006**	**.0038511**	**.0232957**
age	**.0738747**	**.024252**	**3.05**	**0.002**	**.0263416**	**.1214077**
obese_ow						
Yes	**-.765294**	**.4232521**	**-1.81**	**0.071**	**-1.594853**	**.0642649**
sex#c.cholesterol						
Female	**-.0037056**	**.0016851**	**-2.20**	**0.028**	**-.0070083**	**-.0004029**
high_dbp#c.sbp						
No	**.0053884**	**.0020313**	**2.65**	**0.008**	**.001407**	**.0093697**
Yes	**.0067626**	**.0016677**	**4.06**	**0.000**	**.003494**	**.0100312**
c.cholesterol#c.age	**-.0002361**	**.0001007**	**-2.34**	**0.019**	**-.0004335**	**-.0000387**
obese_ow#c.age						
Yes	**.0179147**	**.0086443**	**2.07**	**0.038**	**.0009723**	**.0348572**
_cons	**-13.39632**	**1.219219**	**-10.99**	**0.000**	**-15.78595**	**-11.00669**
ln(persontime)	**1**	(exposure)				
inflate						
sex						
Female	**1.273077**	**.2409671**	**5.28**	**0.000**	**.8007907**	**1.745364**
cholesterol	**-.0097049**	**.002496**	**-3.89**	**0.000**	**-.014597**	**-.0048129**
age	**-.0903298**	**.0125836**	**-7.18**	**0.000**	**-.1149933**	**-.0656663**
_cons	**4.459969**	**.7229926**	**6.17**	**0.000**	**3.04293**	**5.877008**
/lnalpha	**.8772178**	**.0805275**	**10.89**	**0.000**	**.7193868**	**1.035049**
alpha	**2.404202**	**.1936043**			**2.053174**	**2.815244**

```
366 estimates store minussexsbp2

367
368 lrtest (minusdbpage) (minussexsbp2), stats dir
```

Likelihood-ratio test
Assumption: minussexsbp2 nested within minusdbpage

LR chi2(**1**) = **12.14**
Prob > chi2 = **0.0005**

Akaike's information criterion and Bayesian information criterion

Model	N	ll(null)	ll(model)	df	AIC	BIC
minussexsbp2	**4,658**	**-5658.388**	**-5611.895**	**15**	**11253.79**	**11350.48**
minusdbpage	**4,658**	**-5658.388**	**-5605.826**	**16**	**11243.65**	**11346.79**

Note: BIC uses N = number of observations. See **[R] BIC note**.

Name	Command	Dependent variable	Number of param.	Title
minussexsbp2	**zinb**	**cv_events**	**20**	***Zero-inflated negative binomial regression***
minusdbpage	**zinb**	**cv_events**	**22**	***Zero-inflated negative binomial regression***

```
369 *signficant ; keep ; also model with sexsbp has smaller -2LL
370
371
372
373 *DROP  high_dbp#c.sbp
374 zinb cv_events i.sex cholesterol age i.obese_ow i.sex#c.sbp i.sex#c.cholesterol c.ch
  > olesterol#c.age c.age#i.obese_ow, inflate(i.sex cholesterol age) exposure(persontime
  > )

  Fitting constant-only model:

  Iteration 0:   log likelihood =  -6408.826  (not concave)
  Iteration 1:   log likelihood = -5832.2281  (not concave)
  Iteration 2:   log likelihood = -5771.4327  (not concave)
  Iteration 3:   log likelihood = -5757.8771
  Iteration 4:   log likelihood = -5713.0225
  Iteration 5:   log likelihood =  -5665.999
  Iteration 6:   log likelihood = -5659.0027
  Iteration 7:   log likelihood =  -5658.389
  Iteration 8:   log likelihood = -5658.3875
  Iteration 9:   log likelihood = -5658.3875

  Fitting full model:

  Iteration 0:   log likelihood = -5658.3875
  Iteration 1:   log likelihood = -5609.3103
  Iteration 2:   log likelihood = -5606.9733
  Iteration 3:   log likelihood = -5606.9066
  Iteration 4:   log likelihood = -5606.9062

  Zero-inflated negative binomial regression              Number of obs =  4,658
  Inflation model: logit                                  Nonzero obs   =  1,614
                                                          Zero obs      =  3,044
                                                          LR chi2(9)    = 102.96
  Log likelihood = -5606.906                              Prob > chi2   = 0.0000
```

cv_events	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
cv_events						
sex						
Female	2.244675	.5632982	3.98	0.000	1.140631	3.348719
cholesterol	.0133335	.004922	2.71	0.007	.0036866	.0229804
age	.0744902	.0240928	3.09	0.002	.027269	.1217113
obese_ow						
Yes	-.7100452	.4213907	-1.69	0.092	-1.535956	.1158653
sex#c.sbp						
Male	.0142363	.0024699	5.76	0.000	.0093954	.0190771
Female	.003068	.0020069	1.53	0.126	-.0008655	.0070015
sex#c.cholesterol						
Female	-.0030792	.0016944	-1.82	0.069	-.0064002	.0002417
c.cholesterol#c.age	-.0002368	.0001001	-2.37	0.018	-.0004329	-.0000406
obese_ow#c.age						
Yes	.0168888	.0086055	1.96	0.050	.0000222	.0337553
_cons	-14.53585	1.224205	-11.87	0.000	-16.93525	-12.13645
ln(persontime)	1	(exposure)				
inflate						
sex						
Female	1.372769	.2519599	5.45	0.000	.8789368	1.866602
cholesterol	-.0093336	.0024463	-3.82	0.000	-.0141283	-.0045389
age	-.0907157	.0123856	-7.32	0.000	-.1149911	-.0664403
_cons	4.348886	.7120311	6.11	0.000	2.953331	5.744441
/lnalpha	.8610971	.0811086	10.62	0.000	.7021272	1.020067

alpha	**2.365755**	**.191883**	**2.018041**	**2.773381**

```
375 estimates store minusdbpsbp

376
377
378 lrtest (minusdbpage) (minusdbpsbp), stats dir

  Likelihood-ratio test
  Assumption: minusdbpsbp nested within minusdbpage

   LR chi2(1) =   2.16
  Prob > chi2 = 0.1417

  Akaike's information criterion and Bayesian information criterion
```

Model	N	ll(null)	ll(model)	df	AIC	BIC
minusdbpsbp	**4,658**	**-5658.388**	**-5606.906**	**15**	**11243.81**	**11340.51**
minusdbpage	**4,658**	**-5658.388**	**-5605.826**	**16**	**11243.65**	**11346.79**

Note: BIC uses N = number of observations. See **[R] BIC note**.

Name	Command	Dependent variable	Number of param.	Title
minusdbpsbp	**zinb**	**cv_events**	**20**	***Zero-inflated negative binomial regression***
minusdbpage	**zinb**	**cv_events**	**22**	***Zero-inflated negative binomial regression***

```
379 *nonsignificant drop high_dbp#c.sbp
380
381
382 *drop obese_ow
383 zinb cv_events i.sex cholesterol age i.sex#c.sbp i.sex#c.cholesterol c.cholesterol#c
  > .age c.age#i.obese_ow, inflate(i.sex cholesterol age) exposure(persontime)

  Fitting constant-only model:

  Iteration 0:   log likelihood =  -6408.826  (not concave)
  Iteration 1:   log likelihood = -5832.2281  (not concave)
  Iteration 2:   log likelihood = -5771.4327  (not concave)
  Iteration 3:   log likelihood = -5757.8771
  Iteration 4:   log likelihood = -5713.0225
  Iteration 5:   log likelihood =  -5665.999
  Iteration 6:   log likelihood = -5659.0027
  Iteration 7:   log likelihood =  -5658.389
  Iteration 8:   log likelihood = -5658.3875
  Iteration 9:   log likelihood = -5658.3875

  Fitting full model:

  Iteration 0:   log likelihood = -5658.3875
  Iteration 1:   log likelihood = -5610.6255
  Iteration 2:   log likelihood = -5608.3885
  Iteration 3:   log likelihood = -5608.3257
  Iteration 4:   log likelihood = -5608.3253

  Zero-inflated negative binomial regression            Number of obs =  4,658
  Inflation model: logit                                Nonzero obs   =  1,614
                                                        Zero obs      =  3,044
                                                        LR chi2(8)    = 100.12
  Log likelihood = -5608.325                            Prob > chi2   = 0.0000
```

cv_events	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
cv_events						
sex						
Female	**2.275004**	**.5627866**	**4.04**	**0.000**	**1.171963**	**3.378046**
cholesterol	**.0123781**	**.0048975**	**2.53**	**0.011**	**.0027791**	**.0219771**
age	**.0789422**	**.0239826**	**3.29**	**0.001**	**.0319372**	**.1259473**
sex#c.sbp						
Male	**.0141347**	**.0024679**	**5.73**	**0.000**	**.0092977**	**.0189718**
Female	**.0028722**	**.0020021**	**1.43**	**0.151**	**-.0010519**	**.0067964**
sex#c.cholesterol						
Female	**-.0031176**	**.0016977**	**-1.84**	**0.066**	**-.0064451**	**.0002099**
c.cholesterol#c.age	**-.000219**	**.0000997**	**-2.20**	**0.028**	**-.0004144**	**-.0000235**
obese_ow#c.age						
Yes	**.0026039**	**.001515**	**1.72**	**0.086**	**-.0003654**	**.0055733**
_cons	**-14.72577**	**1.220388**	**-12.07**	**0.000**	**-17.11769**	**-12.33386**
ln(persontime)	**1**	(exposure)				
inflate						
sex						
Female	**1.366375**	**.2535454**	**5.39**	**0.000**	**.8694355**	**1.863315**
cholesterol	**-.0093902**	**.0024681**	**-3.80**	**0.000**	**-.0142277**	**-.0045527**
age	**-.0893145**	**.0123346**	**-7.24**	**0.000**	**-.1134899**	**-.065139**
_cons	**4.296386**	**.7152861**	**6.01**	**0.000**	**2.894451**	**5.698321**
/lnalpha	**.8645476**	**.0812149**	**10.65**	**0.000**	**.7053693**	**1.023726**
alpha	**2.373932**	**.1927987**			**2.024594**	**2.783547**

```
384 estimates store minusobese

385
386 lrtest (minusobese) (minusdbpsbp), stats dir
```

Likelihood-ratio test
Assumption: minusobese nested within minusdbpsbp

LR chi2(**1**) = **2.84**
Prob > chi2 = **0.0920**

Akaike's information criterion and Bayesian information criterion

Model	N	ll(null)	ll(model)	df	AIC	BIC
minusobese	**4,658**	**-5658.388**	**-5608.325**	**14**	**11244.65**	**11334.9**
minusdbpsbp	**4,658**	**-5658.388**	**-5606.906**	**15**	**11243.81**	**11340.51**

Note: BIC uses N = number of observations. See **[R] BIC note**.

Name	Command	Dependent variable	Number of param.	Title
minusobese	**zinb**	**cv_events**	**18**	***Zero-inflated negative binomial regression***
minusdbpsbp	**zinb**	**cv_events**	**20**	***Zero-inflated negative binomial regression***

```
387 *nonsignificant drop obese_ow
388
389
390 *drop  obese_ow#c.age
391 zinb cv_events i.sex cholesterol age i.sex#c.sbp i.sex#c.cholesterol c.cholesterol#c
  > .age , inflate(i.sex cholesterol age) exposure(persontime)

  Fitting constant-only model:

  Iteration 0:   log likelihood = -6427.2304  (not concave)
  Iteration 1:   log likelihood = -5848.1257  (not concave)
  Iteration 2:   log likelihood = -5786.8878  (not concave)
  Iteration 3:   log likelihood = -5773.0696
  Iteration 4:   log likelihood = -5727.0294
  Iteration 5:   log likelihood = -5679.6561
  Iteration 6:   log likelihood = -5673.2477
  Iteration 7:   log likelihood = -5672.6691
  Iteration 8:   log likelihood = -5672.6674
  Iteration 9:   log likelihood = -5672.6674

  Fitting full model:

  Iteration 0:   log likelihood = -5672.6674
  Iteration 1:   log likelihood = -5627.1172
  Iteration 2:   log likelihood = -5625.0117
  Iteration 3:   log likelihood =  -5624.949
  Iteration 4:   log likelihood = -5624.9487

  Zero-inflated negative binomial regression              Number of obs =  4,666
  Inflation model: logit                                  Nonzero obs   =  1,617
                                                          Zero obs      =  3,049
                                                          LR chi2(7)    =  95.44
  Log likelihood = -5624.949                              Prob > chi2   = 0.0000
```

cv_events	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
cv_events						
sex						
Female	2.224751	.5648136	3.94	0.000	1.117737	3.331765
cholesterol	.0122626	.0049051	2.50	0.012	.0026487	.0218764
age	.0795956	.0239862	3.32	0.001	.0325836	.1266076
sex#c.sbp						
Male	.0141349	.0024572	5.75	0.000	.0093189	.0189508
Female	.0032908	.0019879	1.66	0.098	-.0006054	.007187
sex#c.cholesterol						
Female	-.0032282	.0017034	-1.90	0.058	-.0065669	.0001105
c.cholesterol#c.age	-.0002109	.0001	-2.11	0.035	-.0004068	-.000015
_cons	-14.73606	1.220474	-12.07	0.000	-17.12815	-12.34398
ln(persontime)	1	(exposure)				
inflate						
sex						
Female	1.372676	.2559785	5.36	0.000	.8709675	1.874385
cholesterol	-.0094586	.0024867	-3.80	0.000	-.0143325	-.0045847
age	-.089443	.0124442	-7.19	0.000	-.1138332	-.0650529
_cons	4.303731	.7196836	5.98	0.000	2.893177	5.714285
/lnalpha	.874418	.0808396	10.82	0.000	.7159752	1.032861
alpha	2.397479	.1938113			2.046181	2.80909

```
392 estimates store minusobeseage

393
394
395 *lrtest (minusobese) (minusobeseage), stats dir
396 *keep model with obeseage as it has smaller -2LL
397
398 *drop sex#c.cholesterol
399 zinb cv_events i.sex cholesterol age i.sex#c.sbp c.cholesterol#c.age c.age#i.obese_o
  > w, inflate(i.sex cholesterol age) exposure(persontime)

  Fitting constant-only model:

  Iteration 0:   log likelihood =  -6408.826  (not concave)
  Iteration 1:   log likelihood = -5832.2281  (not concave)
  Iteration 2:   log likelihood = -5771.4327  (not concave)
  Iteration 3:   log likelihood = -5757.8771
  Iteration 4:   log likelihood = -5713.0225
  Iteration 5:   log likelihood =  -5665.999
  Iteration 6:   log likelihood = -5659.0027
  Iteration 7:   log likelihood =  -5658.389
  Iteration 8:   log likelihood = -5658.3875
  Iteration 9:   log likelihood = -5658.3875

  Fitting full model:

  Iteration 0:   log likelihood = -5658.3875
  Iteration 1:   log likelihood = -5612.0785
  Iteration 2:   log likelihood = -5610.0625
  Iteration 3:   log likelihood =  -5610.021
  Iteration 4:   log likelihood = -5610.0209

  Zero-inflated negative binomial regression              Number of obs =  4,658
  Inflation model: logit                                  Nonzero obs   =  1,614
                                                          Zero obs      =  3,044
                                                          LR chi2(7)    =  96.73
  Log likelihood = -5610.021                              Prob > chi2   = 0.0000
```

cv_events	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
cv_events						
sex						
Female	1.62062	.4349526	3.73	0.000	.7681281	2.473111
cholesterol	.0123749	.004896	2.53	0.011	.0027789	.0219708
age	.0838284	.0238191	3.52	0.000	.0371438	.130513
sex#c.sbp						
Male	.0144801	.0024686	5.87	0.000	.0096416	.0193185
Female	.0026032	.0019947	1.31	0.192	-.0013064	.0065127
c.cholesterol#c.age	-.0002495	.0000983	-2.54	0.011	-.0004422	-.0000568
obese_ow#c.age						
Yes	.0026663	.0015149	1.76	0.078	-.0003028	.0056354
_cons	-14.65856	1.219525	-12.02	0.000	-17.04878	-12.26833
ln(persontime)	1	(exposure)				
inflate						
sex						
Female	1.285732	.2339124	5.50	0.000	.8272722	1.744192
cholesterol	-.0085372	.0023692	-3.60	0.000	-.0131808	-.0038936
age	-.0901857	.012313	-7.32	0.000	-.1143187	-.0660527
_cons	4.227806	.7128123	5.93	0.000	2.830719	5.624892
/lnalpha	.8566664	.0823062	10.41	0.000	.6953492	1.017984
alpha	2.355296	.1938555			2.004409	2.767609

```
400 estimates store minussexcholest

401
402 lrtest (minusobese) (minussexcholest), stats dir

  Likelihood-ratio test
  Assumption: minussexchol~t nested within minusobese

   LR chi2(1) =   3.39
  Prob > chi2 = 0.0655

  Akaike's information criterion and Bayesian information criterion

┌──────────────┬──────────────────────────────────────────────────────────────┐
│        Model │          N   ll(null)  ll(model)      df        AIC        BIC │
├──────────────┼──────────────────────────────────────────────────────────────┤
│ minussexch~t │      4,658  -5658.388  -5610.021      13   11246.04   11329.84 │
│   minusobese │      4,658  -5658.388  -5608.325      14   11244.65    11334.9 │
└──────────────┴──────────────────────────────────────────────────────────────┘
  Note: BIC uses N = number of observations. See [R] BIC note.

┌──────────────┬──────────┬──────────────┬──────────┬──────────────────────────┐
│         Name │ Command  │ Dependent    │ Number of│ Title                    │
│              │          │  variable    │   param. │                          │
├──────────────┼──────────┼──────────────┼──────────┼──────────────────────────┤
│ minussexch~t │ zinb     │ cv_events    │       16 │ Zero-inflated negative   │
│              │          │              │          │ binomial regression      │
│   minusobese │ zinb     │ cv_events    │       18 │ Zero-inflated negative   │
│              │          │              │          │ binomial regression      │
└──────────────┴──────────┴──────────────┴──────────┴──────────────────────────┘

403 *nonsignificant drop
404
405
406 *drop  sex#c.sbp as its most nonsignficant
407 zinb cv_events i.sex cholesterol age c.cholesterol#c.age c.age#i.obese_ow, inflate(i
  > .sex cholesterol age) exposure(persontime)

  Fitting constant-only model:

  Iteration 0:   log likelihood =  -6408.826  (not concave)
  Iteration 1:   log likelihood = -5832.2281  (not concave)
  Iteration 2:   log likelihood = -5771.4327  (not concave)
  Iteration 3:   log likelihood = -5757.8771
  Iteration 4:   log likelihood = -5713.0225
  Iteration 5:   log likelihood =  -5665.999
  Iteration 6:   log likelihood = -5659.0027
  Iteration 7:   log likelihood =  -5658.389
  Iteration 8:   log likelihood = -5658.3875
  Iteration 9:   log likelihood = -5658.3875

  Fitting full model:

  Iteration 0:   log likelihood = -5658.3875
  Iteration 1:   log likelihood = -5629.5256
  Iteration 2:   log likelihood = -5628.5535
  Iteration 3:   log likelihood = -5628.5389
  Iteration 4:   log likelihood = -5628.5389

  Zero-inflated negative binomial regression            Number of obs =  4,658
  Inflation model: logit                                Nonzero obs   =  1,614
                                                        Zero obs      =  3,044
                                                        LR chi2(5)    =  59.70
  Log likelihood = -5628.539                            Prob > chi2   = 0.0000
```

cv_events	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
cv_events						
sex						
Female	-.0032985	.0903601	-0.04	0.971	-.1804011	.1738042
cholesterol	.0131484	.0049329	2.67	0.008	.00348	.0228168
age	.09102	.0239523	3.80	0.000	.0440744	.1379656
c.cholesterol#c.age	-.0002609	.0000992	-2.63	0.009	-.0004554	-.0000665
obese_ow#c.age						
Yes	.0038372	.0015063	2.55	0.011	.000885	.0067895
_cons	-13.112	1.192413	-11.00	0.000	-15.44909	-10.77492
ln(persontime)	1	(exposure)				
inflate						
sex						
Female	1.157938	.2170895	5.33	0.000	.7324503	1.583425
cholesterol	-.0085751	.002358	-3.64	0.000	-.0131967	-.0039534
age	-.0888537	.0122934	-7.23	0.000	-.1129484	-.064759
_cons	4.280679	.7031552	6.09	0.000	2.90252	5.658837
/lnalpha	.8759661	.0838707	10.44	0.000	.7115825	1.04035
alpha	2.401194	.2013899			2.037213	2.830207

```
408
409
410 *drop sex as its most nonsignficant
411 zinb cv_events cholesterol age c.cholesterol#c.age c.age#i.obese_ow, inflate( choles
  > terol age) exposure(persontime)

  Fitting constant-only model:

  Iteration 0:   log likelihood =  -6408.826  (not concave)
  Iteration 1:   log likelihood = -5839.3777  (not concave)
  Iteration 2:   log likelihood = -5806.7868  (not concave)
  Iteration 3:   log likelihood = -5793.1383  (not concave)
  Iteration 4:   log likelihood =  -5785.465  (not concave)
  Iteration 5:   log likelihood = -5776.9236  (not concave)
  Iteration 6:   log likelihood = -5740.3661
  Iteration 7:   log likelihood = -5717.1248
  Iteration 8:   log likelihood = -5689.4376
  Iteration 9:   log likelihood = -5686.6453
  Iteration 10:  log likelihood = -5686.5735
  Iteration 11:  log likelihood = -5686.5734

  Fitting full model:

  Iteration 0:   log likelihood = -5686.5734
  Iteration 1:   log likelihood = -5655.8438
  Iteration 2:   log likelihood = -5654.8181
  Iteration 3:   log likelihood = -5654.8049
  Iteration 4:   log likelihood = -5654.8049

  Zero-inflated negative binomial regression              Number of obs =  4,658
  Inflation model: logit                                  Nonzero obs   =  1,614
                                                          Zero obs      =  3,044
                                                          LR chi2(4)    =  63.54
  Log likelihood = -5654.805                              Prob > chi2   = 0.0000
```

cv_events	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
cv_events						
cholesterol	.0126026	.0049105	2.57	0.010	.0029783	.022227
age	.0900462	.0239182	3.76	0.000	.0431673	.1369251
c.cholesterol#c.age	-.0002547	.0000988	-2.58	0.010	-.0004484	-.000061
obese_ow#c.age						
Yes	.0047099	.0015125	3.11	0.002	.0017454	.0076744
_cons	-13.09627	1.178878	-11.11	0.000	-15.40683	-10.78571
ln(persontime)	1	(exposure)				
inflate						
cholesterol	-.0125335	.0028503	-4.40	0.000	-.0181199	-.0069471
age	-.1006571	.0150304	-6.70	0.000	-.1301162	-.0711981
_cons	6.05261	.8400907	7.20	0.000	4.406063	7.699158
/lnalpha	.9925923	.0795776	12.47	0.000	.836623	1.148562
alpha	2.69822	.214718			2.308558	3.153653

```
412
413
414 *all terms remaining are significant; now bring back interacting predictors
415
416 *obese_ow brought back
417 zinb cv_events cholesterol age i.obese_ow c.cholesterol#c.age c.age#i.obese_ow, infl
  > ate(cholesterol age) exposure(persontime)

  Fitting constant-only model:

  Iteration 0:   log likelihood =  -6408.826  (not concave)
  Iteration 1:   log likelihood = -5839.3777  (not concave)
  Iteration 2:   log likelihood = -5806.7868  (not concave)
  Iteration 3:   log likelihood = -5793.1383  (not concave)
  Iteration 4:   log likelihood =  -5785.465  (not concave)
  Iteration 5:   log likelihood = -5776.9236  (not concave)
  Iteration 6:   log likelihood = -5740.3661
  Iteration 7:   log likelihood = -5717.1248
  Iteration 8:   log likelihood = -5689.4376
  Iteration 9:   log likelihood = -5686.6453
  Iteration 10:  log likelihood = -5686.5735
  Iteration 11:  log likelihood = -5686.5734

  Fitting full model:

  Iteration 0:   log likelihood = -5686.5734
  Iteration 1:   log likelihood =  -5655.572
  Iteration 2:   log likelihood = -5654.5181
  Iteration 3:   log likelihood = -5654.5056
  Iteration 4:   log likelihood = -5654.5056

  Zero-inflated negative binomial regression              Number of obs =  4,658
  Inflation model: logit                                  Nonzero obs   =  1,614
                                                          Zero obs      =  3,044
                                                          LR chi2(5)    =  64.14
  Log likelihood = -5654.506                              Prob > chi2   = 0.0000
```

cv_events	Coefficient	Std. err.	z	P>|z|	[95% conf.	interval]
cv_events						
cholesterol	.0130951	.0049491	2.65	0.008	.003395	.0227952
age	.0882956	.0240119	3.68	0.000	.0412331	.1353581
obese_ow						
Yes	-.3317215	.4286687	-0.77	0.439	-1.171897	.5084537
c.cholesterol#c.age	-.000264	.0000995	-2.65	0.008	-.000459	-.000069
obese_ow#c.age						
Yes	.0114028	.0087775	1.30	0.194	-.0058007	.0286063
_cons	-13.01788	1.182856	-11.01	0.000	-15.33624	-10.69953
ln(persontime)	1	(exposure)				
inflate						
cholesterol	-.0125108	.002836	-4.41	0.000	-.0180693	-.0069523
age	-.1011032	.0150196	-6.73	0.000	-.130541	-.0716654
_cons	6.075675	.8381821	7.25	0.000	4.432869	7.718482
/lnalpha	.9901042	.0797768	12.41	0.000	.8337445	1.146464
alpha	2.691515	.2147205			2.301922	3.147045

```
418
419 * obese_ow#c.age  no longer signficant; we drop both_cat
420
421 *final model after interaction(crude)
422 zinb cv_events cholesterol age c.cholesterol#c.age, inflate(cholesterol age) exposur
  > e(persontime) irr

  Fitting constant-only model:

  Iteration 0:   log likelihood = -6427.2304  (not concave)
  Iteration 1:   log likelihood = -5855.2434  (not concave)
  Iteration 2:   log likelihood = -5822.1737  (not concave)
  Iteration 3:   log likelihood = -5808.2837  (not concave)
  Iteration 4:   log likelihood = -5800.1138  (not concave)
  Iteration 5:   log likelihood = -5791.2359  (not concave)
  Iteration 6:   log likelihood = -5773.0223  (not concave)
  Iteration 7:   log likelihood =  -5763.651
  Iteration 8:   log likelihood = -5731.2697
  Iteration 9:   log likelihood = -5704.3235
  Iteration 10:  log likelihood = -5701.0397
  Iteration 11:  log likelihood = -5701.0099
  Iteration 12:  log likelihood = -5701.0098

  Fitting full model:

  Iteration 0:   log likelihood = -5701.0098
  Iteration 1:   log likelihood = -5674.7356
  Iteration 2:   log likelihood = -5673.8642
  Iteration 3:   log likelihood = -5673.8533
  Iteration 4:   log likelihood = -5673.8533

  Zero-inflated negative binomial regression              Number of obs =  4,666
  Inflation model: logit                                  Nonzero obs   =  1,617
                                                          Zero obs      =  3,049
                                                          LR chi2(3)    =  54.31
  Log likelihood = -5673.853                              Prob > chi2   = 0.0000
```

cv_events	IRR	Std. err.	z	P>|z|	[95% conf.	interval]
cv_events						
cholesterol	**1.013102**	**.0049865**	**2.64**	**0.008**	**1.003376**	**1.022923**
age	**1.099497**	**.02629**	**3.97**	**0.000**	**1.049158**	**1.15225**
c.cholesterol#c.age	**.9997433**	**.0000992**	**-2.59**	**0.010**	**.999549**	**.9999376**
_cons	**1.74e-06**	**2.05e-06**	**-11.24**	**0.000**	**1.72e-07**	**.0000176**
ln(persontime)	**1**	(exposure)				
inflate						
cholesterol	**-.0125256**	**.0028458**	**-4.40**	**0.000**	**-.0181033**	**-.006948**
age	**-.1004053**	**.0150304**	**-6.68**	**0.000**	**-.1298643**	**-.0709463**
_cons	**6.049216**	**.8402464**	**7.20**	**0.000**	**4.402363**	**7.696068**
/lnalpha	**.9998832**	**.0797781**	**12.53**	**0.000**	**.8435209**	**1.156245**
alpha	**2.717964**	**.2168342**			**2.324537**	**3.177979**

Note: Estimates are transformed only in the first equation to incidence-rate ratios.
Note: **_cons** estimates baseline incidence rate.

```
423
424
425 *CHECK FOR CONFOUNDING;
426 *BRING in this order; sex, obese_ow, sbp dbp
427
428 zinb cv_events i.sex cholesterol age c.cholesterol#c.age, inflate(i.sex cholesterol
  > age) exposure(persontime) irr

  Fitting constant-only model:

  Iteration 0:   log likelihood = -6427.2304  (not concave)
  Iteration 1:   log likelihood = -5848.1257  (not concave)
  Iteration 2:   log likelihood = -5786.8878  (not concave)
  Iteration 3:   log likelihood = -5773.0696
  Iteration 4:   log likelihood = -5727.0294
  Iteration 5:   log likelihood = -5679.6561
  Iteration 6:   log likelihood = -5673.2477
  Iteration 7:   log likelihood = -5672.6691
  Iteration 8:   log likelihood = -5672.6674
  Iteration 9:   log likelihood = -5672.6674

  Fitting full model:

  Iteration 0:   log likelihood = -5672.6674
  Iteration 1:   log likelihood = -5646.7427
  Iteration 2:   log likelihood = -5645.8521
  Iteration 3:   log likelihood = -5645.8389
  Iteration 4:   log likelihood = -5645.8389

  Zero-inflated negative binomial regression              Number of obs =  4,666
  Inflation model: logit                                  Nonzero obs   =  1,617
                                                          Zero obs      =  3,049
                                                          LR chi2(4)    =  53.66
  Log likelihood = -5645.839                              Prob > chi2   = 0.0000
```

cv_events	IRR	Std. err.	z	P>|z|	[95% conf.	interval]
cv_events						
sex						
Female	**.9743113**	**.0880374**	**-0.29**	**0.773**	**.8161771**	**1.163084**
cholesterol	**1.013262**	**.0050115**	**2.66**	**0.008**	**1.003487**	**1.023132**
age	**1.097899**	**.0263075**	**3.90**	**0.000**	**1.047529**	**1.150691**
c.cholesterol#c.age	**.9997449**	**.0000995**	**-2.56**	**0.010**	**.9995499**	**.9999401**
_cons	**1.90e-06**	**2.27e-06**	**-11.03**	**0.000**	**1.83e-07**	**.0000197**
ln(persontime)	**1**	(exposure)				

inflate						
sex						
Female	**1.166517**	**.2186056**	**5.34**	**0.000**	**.7380576**	**1.594976**
cholesterol	**-.008568**	**.0023664**	**-3.62**	**0.000**	**-.0132061**	**-.00393**
age	**-.088998**	**.0123836**	**-7.19**	**0.000**	**-.1132693**	**-.0647267**
_cons	**4.276393**	**.705446**	**6.06**	**0.000**	**2.893744**	**5.659042**
/lnalpha	**.8855678**	**.0837023**	**10.58**	**0.000**	**.7215144**	**1.049621**
alpha	**2.424361**	**.2029245**			**2.057547**	**2.856569**

Note: Estimates are transformed only in the first equation to incidence-rate ratios.
Note: **_cons** estimates baseline incidence rate.

```
429 *no percent chnage >10% in any of the preditors; so sex not confounder
430
431
432 *return obese_ow
433 zinb cv_events cholesterol age c.cholesterol#c.age i.obese_ow, inflate(cholesterol a
  > ge) exposure(persontime) irr

  Fitting constant-only model:

  Iteration 0:   log likelihood =  -6408.826  (not concave)
  Iteration 1:   log likelihood = -5839.3777  (not concave)
  Iteration 2:   log likelihood = -5806.7868  (not concave)
  Iteration 3:   log likelihood = -5793.1383  (not concave)
  Iteration 4:   log likelihood =  -5785.465  (not concave)
  Iteration 5:   log likelihood = -5776.9236  (not concave)
  Iteration 6:   log likelihood = -5740.3661
  Iteration 7:   log likelihood = -5717.1248
  Iteration 8:   log likelihood = -5689.4376
  Iteration 9:   log likelihood = -5686.6453
  Iteration 10:  log likelihood = -5686.5735
  Iteration 11:  log likelihood = -5686.5734

  Fitting full model:

  Iteration 0:   log likelihood = -5686.5734
  Iteration 1:   log likelihood = -5656.3832
  Iteration 2:   log likelihood =  -5655.361
  Iteration 3:   log likelihood = -5655.3481
  Iteration 4:   log likelihood =  -5655.348

  Zero-inflated negative binomial regression              Number of obs =  4,658
  Inflation model: logit                                  Nonzero obs   =  1,614
                                                          Zero obs      =  3,044
                                                          LR chi2(4)    =  62.45
  Log likelihood = -5655.348                              Prob > chi2   = 0.0000
```

cv_events	IRR	Std. err.	z	P>|z|	[95% conf.	interval]
cv_events						
cholesterol	**1.01241**	**.0049809**	**2.51**	**0.012**	**1.002694**	**1.022219**
age	**1.095966**	**.0261969**	**3.83**	**0.000**	**1.045805**	**1.148532**
c.cholesterol#c.age	**.9997507**	**.000099**	**-2.52**	**0.012**	**.9995568**	**.9999447**
obese_ow						
Yes	**1.24178**	**.091788**	**2.93**	**0.003**	**1.074303**	**1.435366**
_cons	**1.92e-06**	**2.26e-06**	**-11.17**	**0.000**	**1.91e-07**	**.0000193**
ln(persontime)	**1**	(exposure)				
inflate						
cholesterol	**-.012543**	**.0028586**	**-4.39**	**0.000**	**-.0181457**	**-.0069403**
age	**-.1003657**	**.0150412**	**-6.67**	**0.000**	**-.1298458**	**-.0708856**	
_cons	**6.037455**	**.8415426**	**7.17**	**0.000**	**4.388062**	**7.686849**
/lnalpha	**.9943716**	**.0795191**	**12.50**	**0.000**	**.838517**	**1.150226**

alpha	**2.703025**	**.2149422**		**2.312934**	**3.158907**

Note: Estimates are transformed only in the first equation to incidence-rate ratios.
Note: **_cons** estimates baseline incidence rate.

```
434 *obese_ow not confounder
435
436 *return sbp
437 zinb cv_events i.sex cholesterol age c.cholesterol#c.age sbp , inflate(i.sex cholest
  > erol age sbp ) exposure(persontime) irr

Fitting constant-only model:

Iteration 0:   log likelihood = -6427.2304  (not concave)
Iteration 1:   log likelihood = -5844.8094  (not concave)
Iteration 2:   log likelihood =  -5780.647  (not concave)
Iteration 3:   log likelihood = -5765.5714  (not concave)
Iteration 4:   log likelihood = -5758.1478  (not concave)
Iteration 5:   log likelihood = -5745.1496  (not concave)
Iteration 6:   log likelihood = -5738.1322  (not concave)
Iteration 7:   log likelihood = -5733.1381  (not concave)
Iteration 8:   log likelihood = -5728.9115  (not concave)
Iteration 9:   log likelihood = -5724.4589  (not concave)
Iteration 10:  log likelihood = -5720.4773  (not concave)
Iteration 11:  log likelihood = -5716.5978  (not concave)
Iteration 12:  log likelihood = -5712.6816  (not concave)
Iteration 13:  log likelihood = -5709.0259
Iteration 14:  log likelihood = -5687.6064
Iteration 15:  log likelihood = -5651.8647
Iteration 16:  log likelihood = -5648.6605
Iteration 17:  log likelihood = -5648.4852
Iteration 18:  log likelihood = -5648.4844
Iteration 19:  log likelihood = -5648.4844

Fitting full model:

Iteration 0:   log likelihood = -5648.4844
Iteration 1:   log likelihood = -5620.0429
Iteration 2:   log likelihood = -5619.0796
Iteration 3:   log likelihood = -5619.0686
Iteration 4:   log likelihood = -5619.0686

Zero-inflated negative binomial regression            Number of obs =  4,666
Inflation model: logit                                Nonzero obs   =  1,617
                                                      Zero obs      =  3,049
                                                      LR chi2(5)    =  58.83
Log likelihood = -5619.069                            Prob > chi2   = 0.0000
```

cv_events	IRR	Std. err.	z	P>|z|	[95% conf.	interval]
cv_events						
sex						
Female	**.9666104**	**.0852459**	**-0.39**	**0.700**	**.8131741**	**1.148998**
cholesterol	**1.013357**	**.004893**	**2.75**	**0.006**	**1.003812**	**1.022993**
age	**1.094826**	**.0257423**	**3.85**	**0.000**	**1.045517**	**1.146461**
c.cholesterol#c.age	**.9997391**	**.0000972**	**-2.68**	**0.007**	**.9995485**	**.9999296**
sbp	**1.003493**	**.0016871**	**2.07**	**0.038**	**1.000192**	**1.006805**
_cons	**1.47e-06**	**1.71e-06**	**-11.49**	**0.000**	**1.48e-07**	**.0000145**
ln(persontime)	**1**	(exposure)				
inflate						
sex						
Female	**1.006468**	**.1866364**	**5.39**	**0.000**	**.6406673**	**1.372269**
cholesterol	**-.0065644**	**.0020959**	**-3.13**	**0.002**	**-.0106723**	**-.0024566**
age	**-.0620846**	**.0115626**	**-5.37**	**0.000**	**-.0847468**	**-.0394223**
sbp	**-.023195**	**.0047058**	**-4.93**	**0.000**	**-.0324183**	**-.0139717**
_cons	**5.905543**	**.7772224**	**7.60**	**0.000**	**4.382215**	**7.428871**
/lnalpha	**.8036418**	**.0855719**	**9.39**	**0.000**	**.6359239**	**.9713597**

alpha	**2.233661**	**.1911386**		**1.888766**	**2.641534**

Note: Estimates are transformed only in the first equation to incidence-rate ratios.
Note: **_cons** estimates baseline incidence rate.

```
438 *sbp not confounder
439
440
441 *return dbp
442 zinb cv_events cholesterol age c.cholesterol#c.age i.high_dbp, inflate(cholesterol a
  > ge) exposure(persontime) irr

  Fitting constant-only model:

  Iteration 0:   log likelihood = -6427.2304  (not concave)
  Iteration 1:   log likelihood = -5855.2434  (not concave)
  Iteration 2:   log likelihood = -5822.1737  (not concave)
  Iteration 3:   log likelihood = -5808.2837  (not concave)
  Iteration 4:   log likelihood = -5800.1138  (not concave)
  Iteration 5:   log likelihood = -5791.2359  (not concave)
  Iteration 6:   log likelihood = -5773.0223  (not concave)
  Iteration 7:   log likelihood =  -5763.651
  Iteration 8:   log likelihood = -5731.2697
  Iteration 9:   log likelihood = -5704.3235
  Iteration 10:  log likelihood = -5701.0397
  Iteration 11:  log likelihood = -5701.0099
  Iteration 12:  log likelihood = -5701.0098

  Fitting full model:

  Iteration 0:   log likelihood = -5701.0098
  Iteration 1:   log likelihood = -5665.4045
  Iteration 2:   log likelihood =  -5664.086
  Iteration 3:   log likelihood = -5664.0658
  Iteration 4:   log likelihood = -5664.0657

  Zero-inflated negative binomial regression              Number of obs =  4,666
  Inflation model: logit                                  Nonzero obs   =  1,617
                                                          Zero obs      =  3,049
                                                          LR chi2(4)    =  73.89
  Log likelihood = -5664.066                              Prob > chi2   = 0.0000
```

cv_events	IRR	Std. err.	z	P>|z|	[95% conf.	interval]
cv_events						
cholesterol	**1.012093**	**.0049786**	**2.44**	**0.015**	**1.002382**	**1.021898**
age	**1.092827**	**.026133**	**3.71**	**0.000**	**1.042789**	**1.145266**
c.cholesterol#c.age	**.9997609**	**.0000991**	**-2.41**	**0.016**	**.9995667**	**.9999551**
high_dbp						
Yes	**1.606153**	**.1764191**	**4.31**	**0.000**	**1.295064**	**1.991968**
_cons	**2.24e-06**	**2.65e-06**	**-11.02**	**0.000**	**2.22e-07**	**.0000227**
ln(persontime)	**1**	(exposure)				
inflate						
cholesterol	**-.012923**	**.0029033**	**-4.45**	**0.000**	**-.0186133**	**-.0072327**
age	**-.1020379**	**.0153217**	**-6.66**	**0.000**	**-.132068**	**-.0720079**
_cons	**6.163653**	**.854548**	**7.21**	**0.000**	**4.48877**	**7.838537**
/lnalpha	**.9988639**	**.078061**	**12.80**	**0.000**	**.8458671**	**1.151861**
alpha	**2.715195**	**.2119509**			**2.329997**	**3.164075**

Note: Estimates are transformed only in the first equation to incidence-rate ratios.
Note: **_cons** estimates baseline incidence rate.

```
443 *dbp not confounder
444
445
446 *FINAL MODEL
447 zinb cv_events cholesterol age c.cholesterol#c.age, inflate(cholesterol age) exposur
  > e(persontime) irr

  Fitting constant-only model:

  Iteration 0:   log likelihood = -6427.2304  (not concave)
  Iteration 1:   log likelihood = -5855.2434  (not concave)
  Iteration 2:   log likelihood = -5822.1737  (not concave)
  Iteration 3:   log likelihood = -5808.2837  (not concave)
  Iteration 4:   log likelihood = -5800.1138  (not concave)
  Iteration 5:   log likelihood = -5791.2359  (not concave)
  Iteration 6:   log likelihood = -5773.0223  (not concave)
  Iteration 7:   log likelihood =  -5763.651
  Iteration 8:   log likelihood = -5731.2697
  Iteration 9:   log likelihood = -5704.3235
  Iteration 10:  log likelihood = -5701.0397
  Iteration 11:  log likelihood = -5701.0099
  Iteration 12:  log likelihood = -5701.0098

  Fitting full model:

  Iteration 0:   log likelihood = -5701.0098
  Iteration 1:   log likelihood = -5674.7356
  Iteration 2:   log likelihood = -5673.8642
  Iteration 3:   log likelihood = -5673.8533
  Iteration 4:   log likelihood = -5673.8533

  Zero-inflated negative binomial regression              Number of obs =  4,666
  Inflation model: logit                                  Nonzero obs   =  1,617
                                                          Zero obs      =  3,049
                                                          LR chi2(3)    =  54.31
  Log likelihood = -5673.853                              Prob > chi2   = 0.0000
```

cv_events	IRR	Std. err.	z	P>|z|	[95% conf.	interval]
cv_events						
cholesterol	1.013102	.0049865	2.64	0.008	1.003376	1.022923
age	1.099497	.02629	3.97	0.000	1.049158	1.15225
c.cholesterol#c.age	.9997433	.0000992	-2.59	0.010	.999549	.9999376
_cons	1.74e-06	2.05e-06	-11.24	0.000	1.72e-07	.0000176
ln(persontime)	1	(exposure)				
inflate						
cholesterol	-.0125256	.0028458	-4.40	0.000	-.0181033	-.006948
age	-.1004053	.0150304	-6.68	0.000	-.1298643	-.0709463
_cons	6.049216	.8402464	7.20	0.000	4.402363	7.696068
/lnalpha	.9998832	.0797781	12.53	0.000	.8435209	1.156245
alpha	2.717964	.2168342			2.324537	3.177979

Note: Estimates are transformed only in the first equation to incidence-rate ratios.
Note: **_cons** estimates baseline incidence rate.

```
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449
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465
466
467
468
469 log close
       name:  <unnamed>
        log:  E:\Documents\STATA\cardiovascular.smcl
   log type:  smcl
  closed on:  16 May 2026, 21:06:32
```